Digital manufacturing - management engineering

Work project

Design, simulation and manufacturing of a gameboy

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Summary

01 Engineering design lab

02 Industrialization lab

03 Manufacturing lab

Engineering design lab

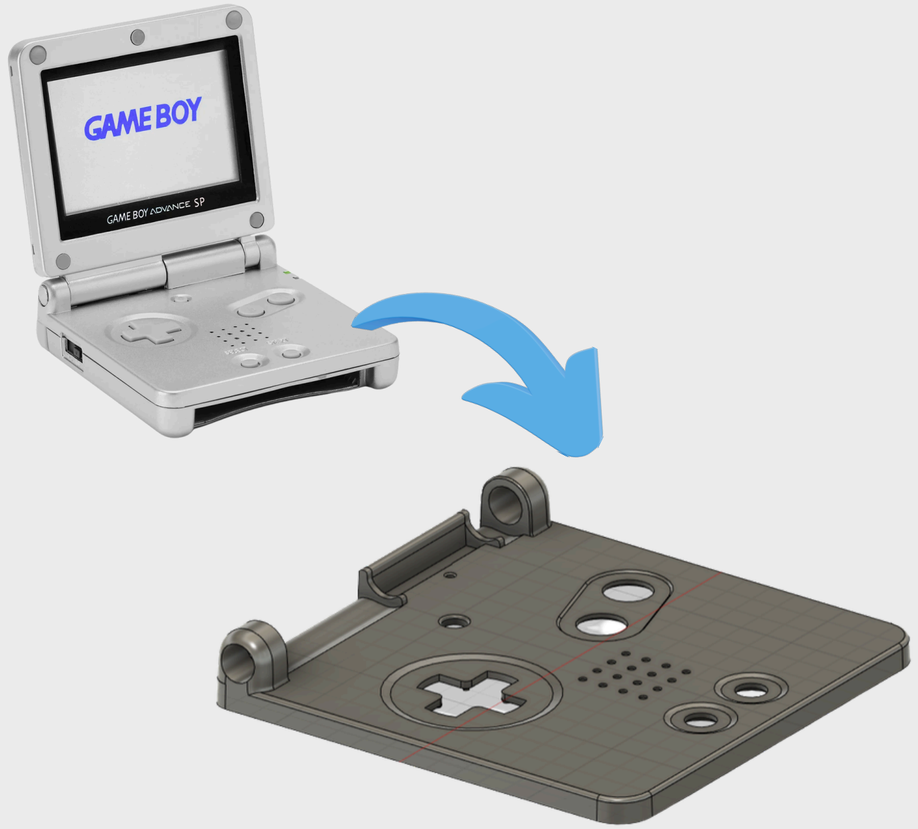
Engineering Design Lab Activity

Game Boy Advance SP Upper Case

01 Reverse engineering
We conducted the reverse engineering activity from the real GBA SP case in this project

02 Parametric Design
All the parameters in this design are parametric to make sure it’s flexible and aligned with all dimensional aspects

03 Load case
Identification of average and extreme operating conditions. Additionally, a maximum displacement will be defined.

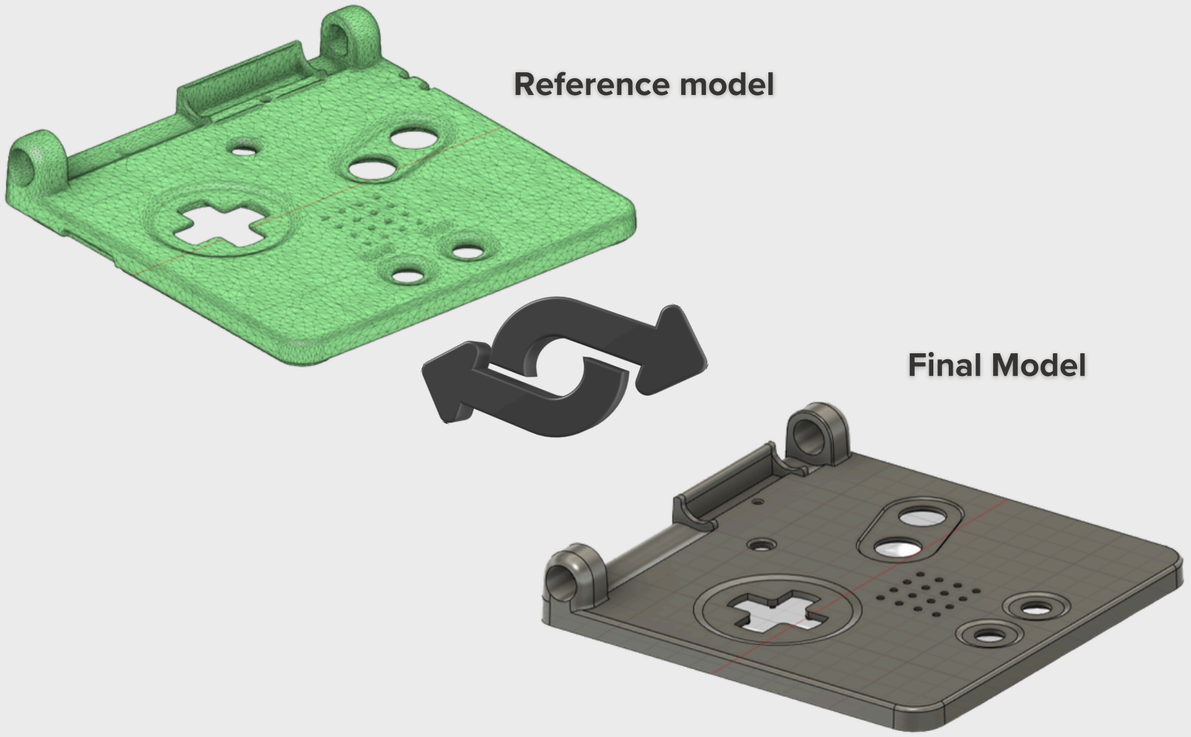


04 Material Selection
Material selection activity was also conducted to ensure our design meets the functional aspecs by using specific materials.

05 Stress Analysis
For this activity we tried to simulate the performance of our model in terms of stress distribution to make sure it complies with normal utilization.

06 Design Optimization
Based on the test result, we tried to improve our design to minimize deflections by varying thickness and ribs applications.

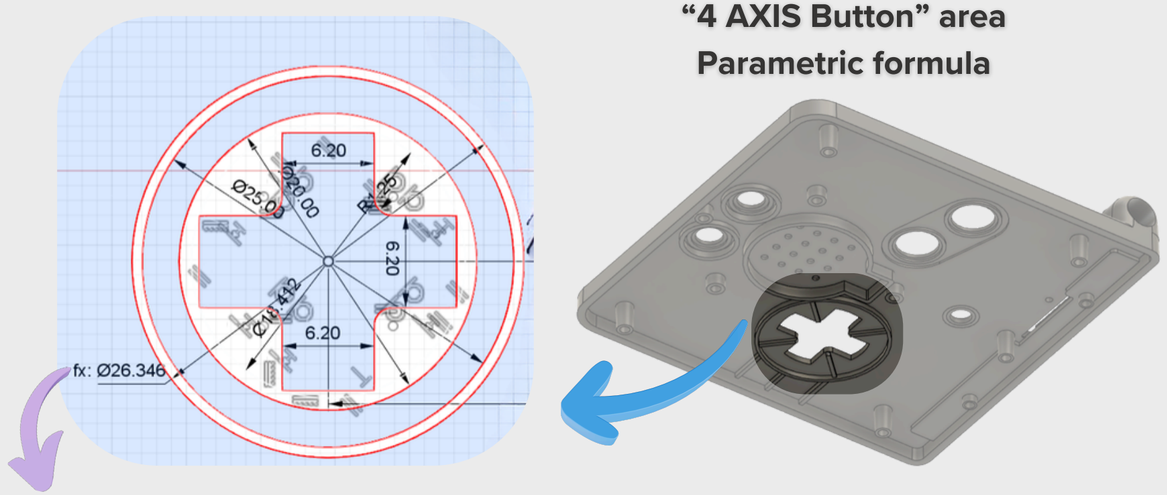
1. Reverse Engineering



We created our model with the real dimensions of the GBA case. We tried to maintain the ratio of the dimensions maintaining it’s original design such as button holes, hinge, speaker holes, screw bosses.

2. Parametric Design

“4 AXIS Button” area
Parametric formula



2 * (-thickness_shell / 1.75) * 0.736 + d136 fx

On the formula above we try to make it align with shell thickness & dimension ratio of the reference dimension

Our model also applied parametric design aligned with specific value as reference. For our case, we took the main thickness as our major reference. With this action, we could have flexible and accurate dimensions when we conduct improvements for example increasing thickness.

3. Load conditions and constraints

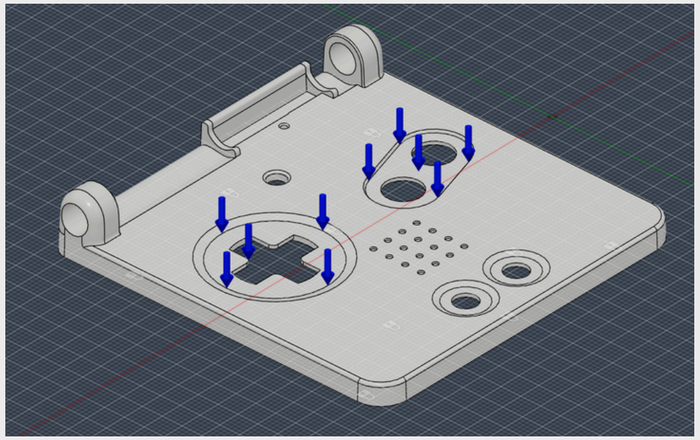
According to some standards:

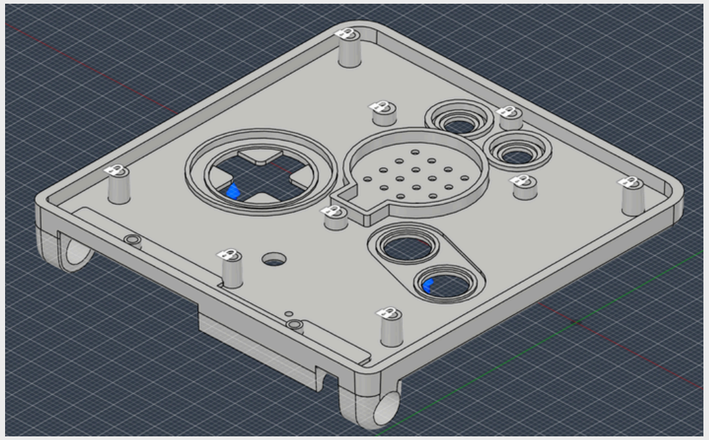
- Nominal force: up to 5N
- Extreme force: 10N
- Maximum displacement: 0,1mm
  (Not main qc item)

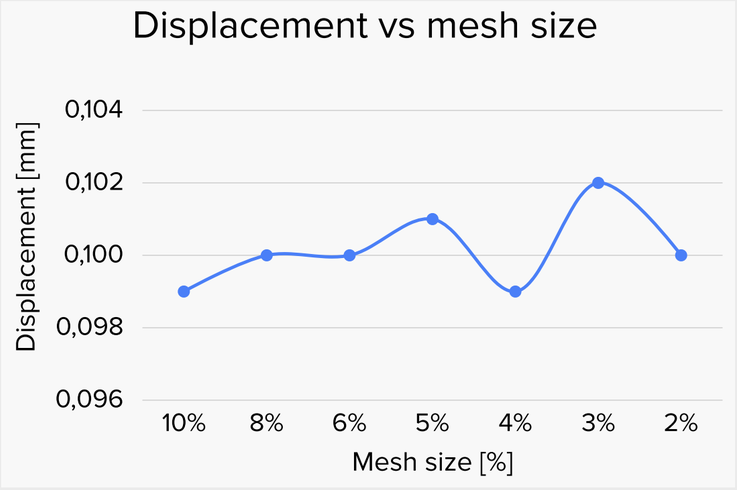
Our load condition:

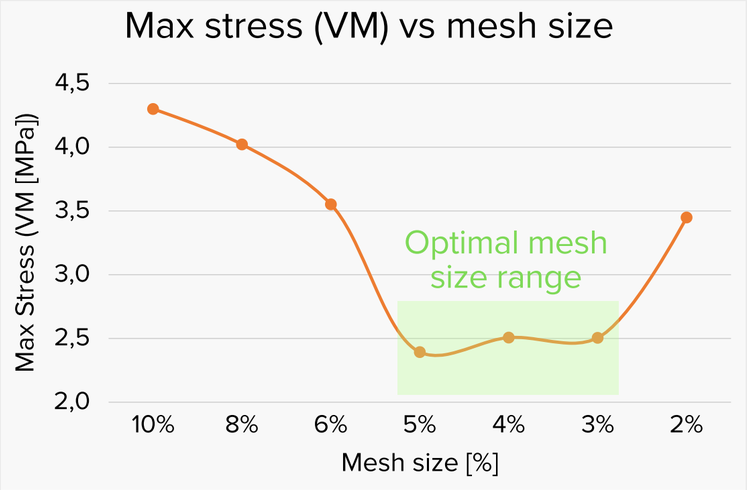
- Pressing both buttons with 10N
- Constraints on bosses

Mesh sensitivity analysis: 4%









Engineering design lab

4. Material selection

01 ABS → reference material, ductile (E=2,24 GPa)

02 PC → tough, more expensive (E=2,275 GPa)

03 PP → less performant, cheaper (E=1,34 GPa)

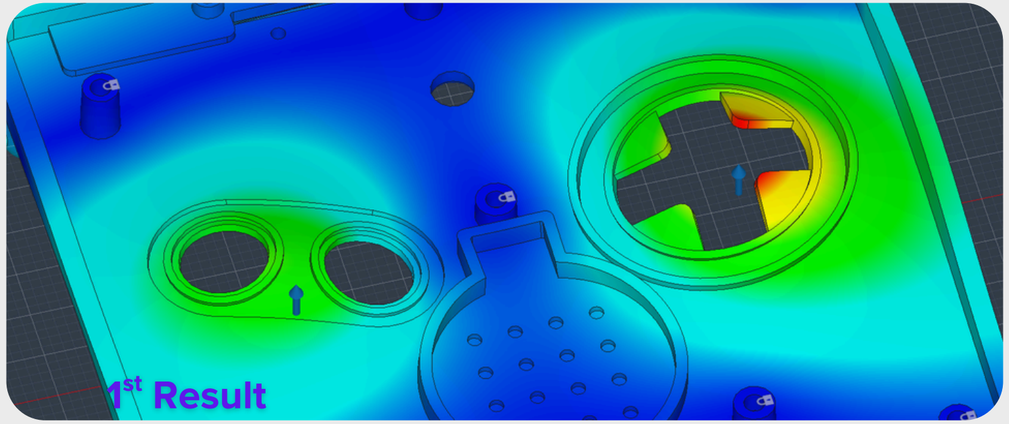
Objectives:

- Minimizing the thickness
- Minimizing the cost

Results (one cavity, injection molding, 1M parts):

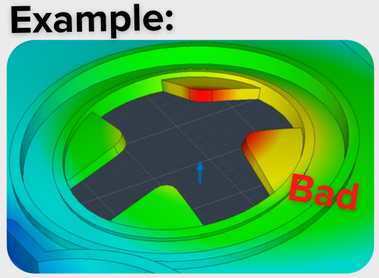
- **ABS: t=2mm → V=15,35 cm^3 → c=0,345 $/part**
- PC: t=2mm → V=15,35 cm^3 → c=0,378 $/part
- PP: t=2,6mm → V=19,45 cm^3 → c=0,538 $/part

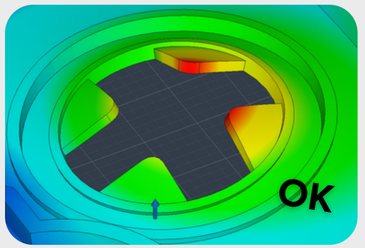
5. Stress analysis



DISPLACEMENT ISSUE FOUND ON 4 AXIS BUTTON AREA
THE DISPLACEMENT VALUE REACHES 0.102 (mm) LOCALIZED

Example:



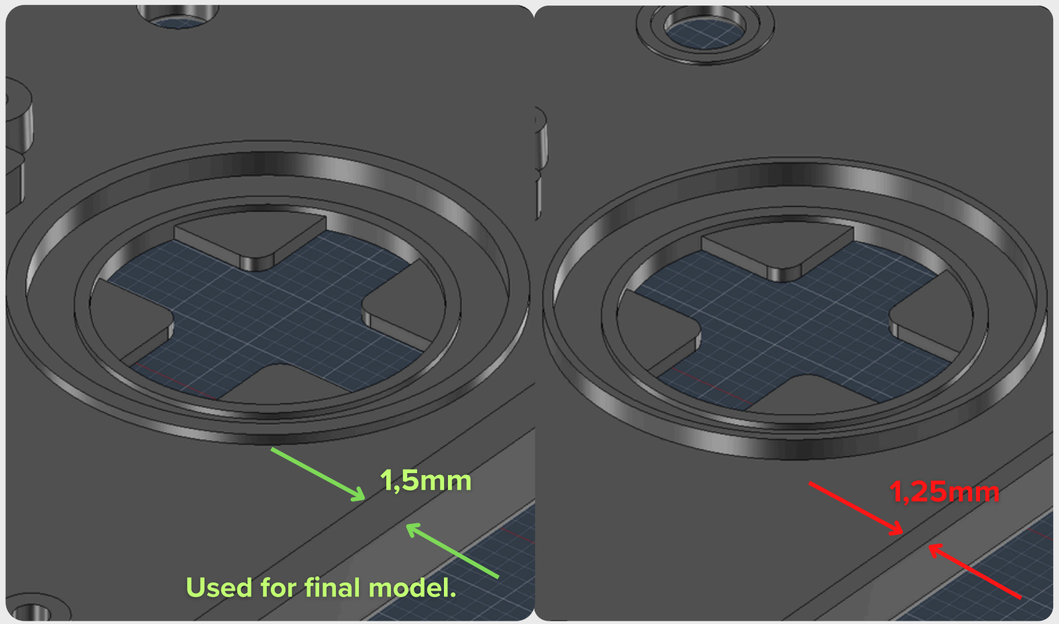


Normal deflection should be below 01mm , it means our model could withstand daily usage of button pressing

6. Design Optimization

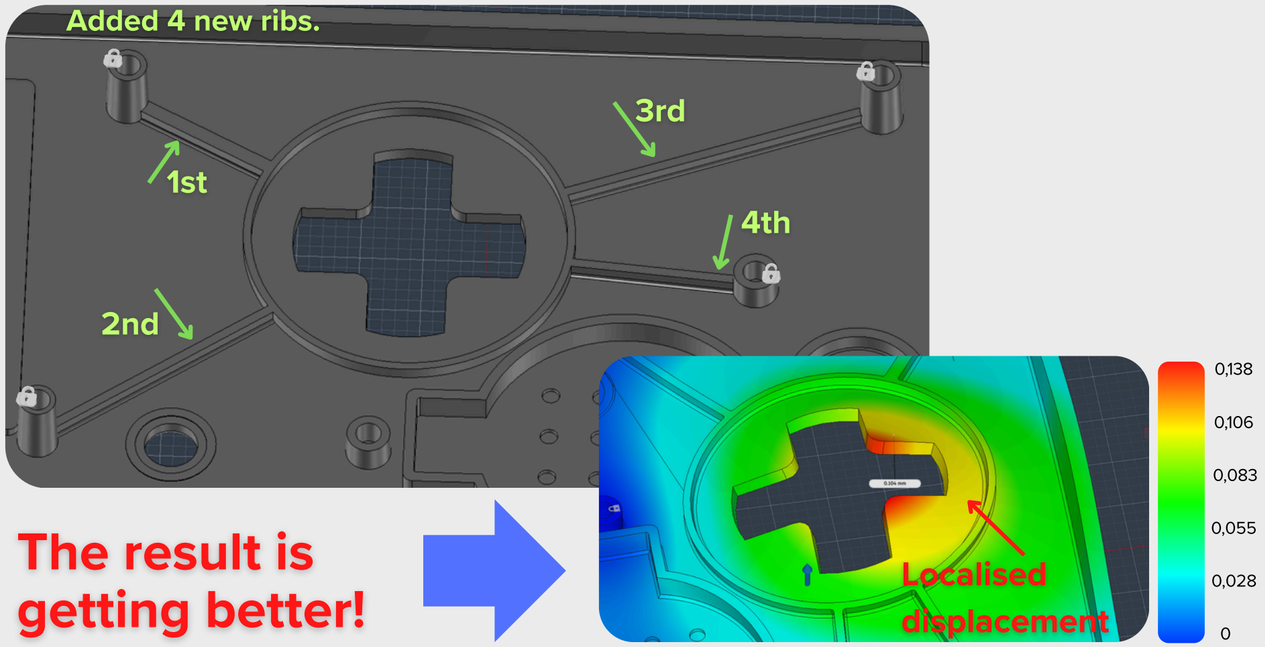
Based on the result, we got a bad result for the displacement analysis around the button area. We tried to optimize with 2 methods.

1. Thickness variation



We tested some variations of the overall thickness (1,5mm & 1,25mm) constrained by the cost while solving the displacement issue.

2. Additional Ribs

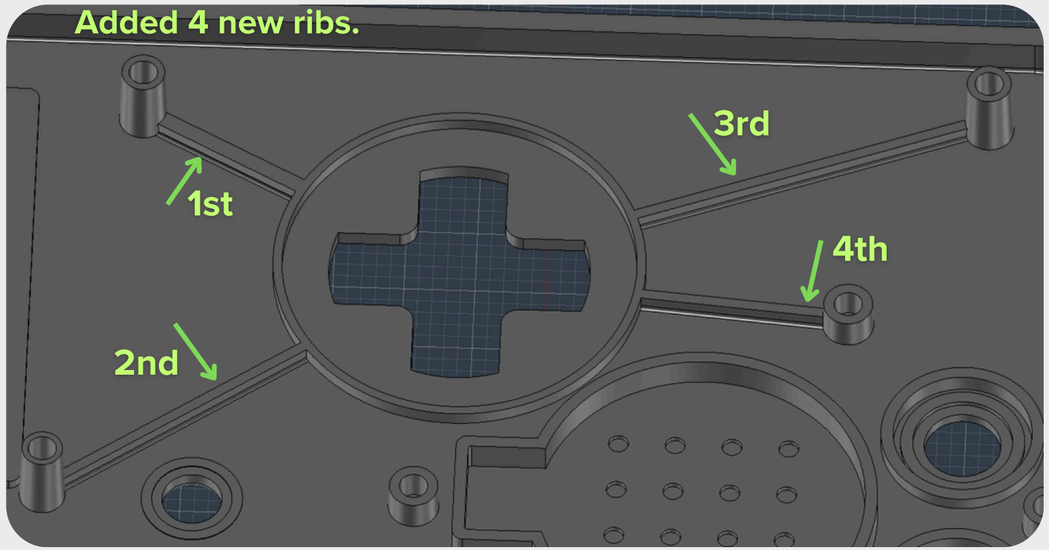


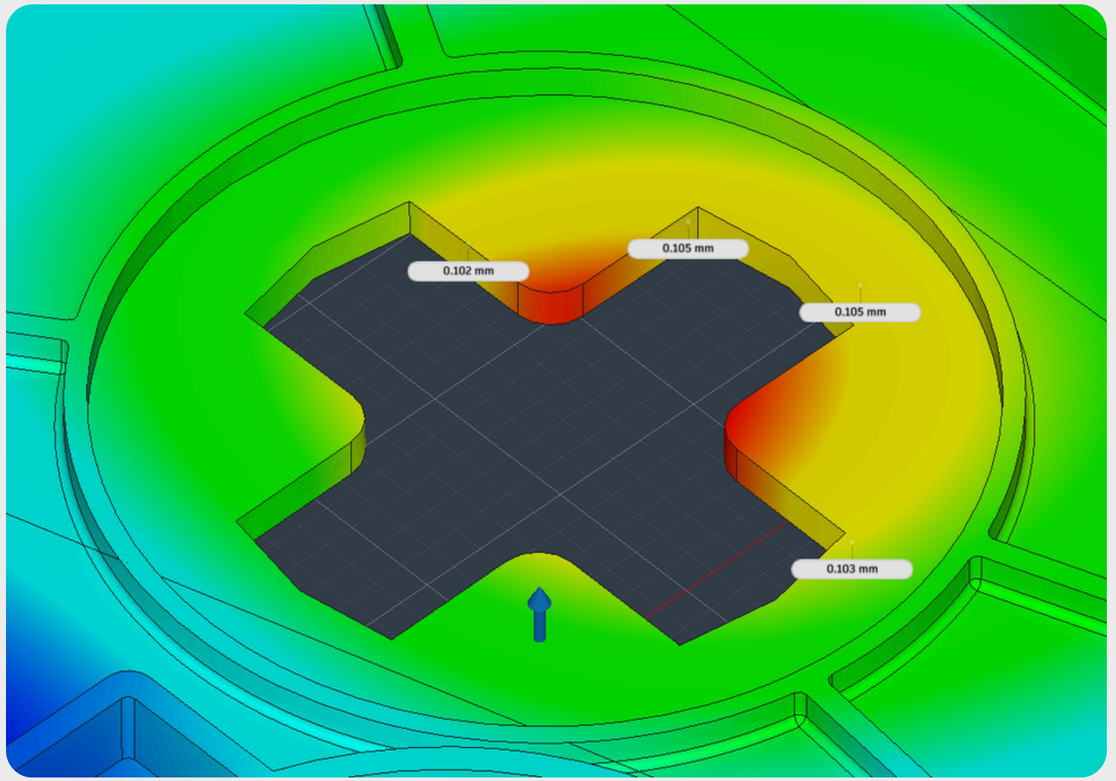
We implemented additional ribs around stress affected area to improve displacement distribution. With this optimization we could solve the problem since general displacement is under threshold.

6. Design Optimization

Based on the result, we got a bad result for the stress analysis around the button area.

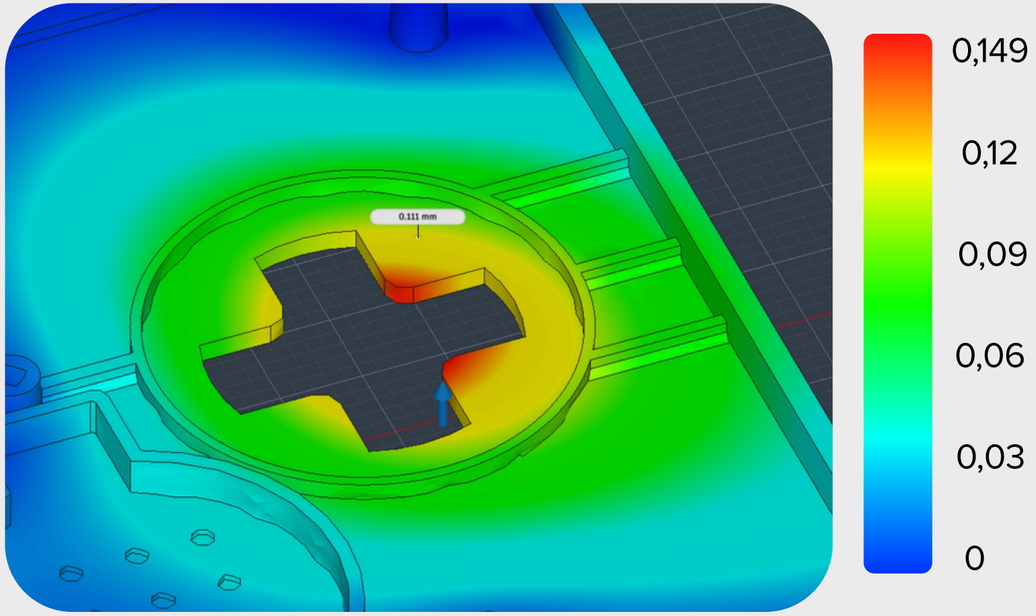
2. Additional Ribs

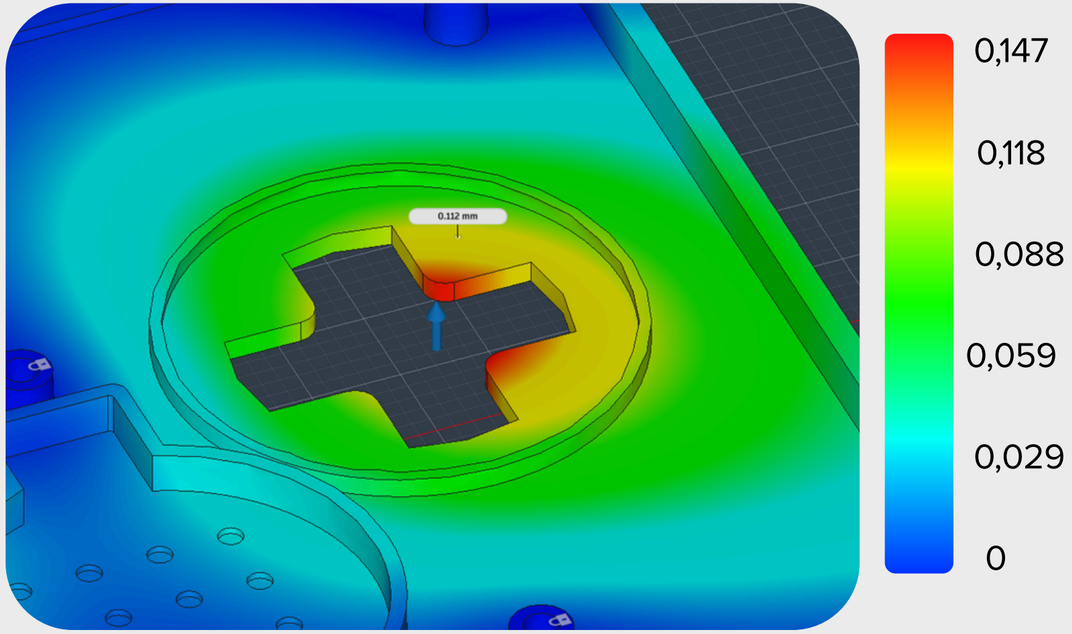




6. Design Optimization - trial and error for ribs

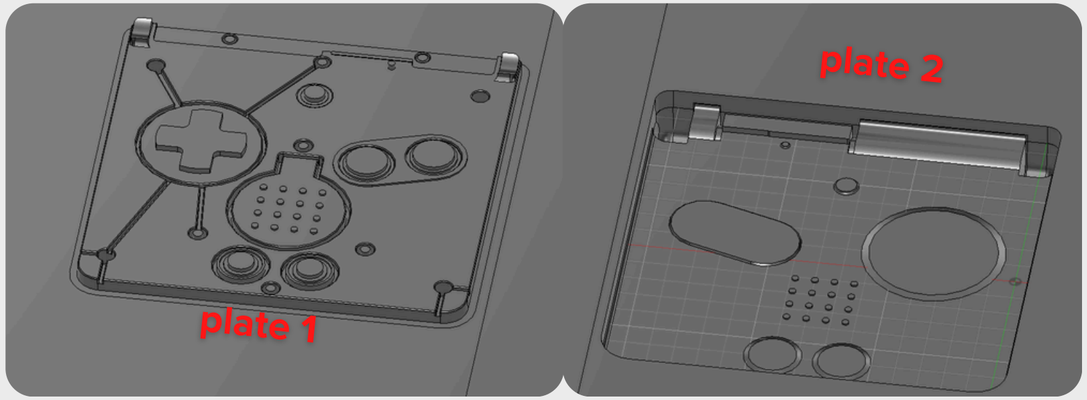
Somehow it’s even worse with the ribs





Injection molding simulation

2 Plate Mold Design

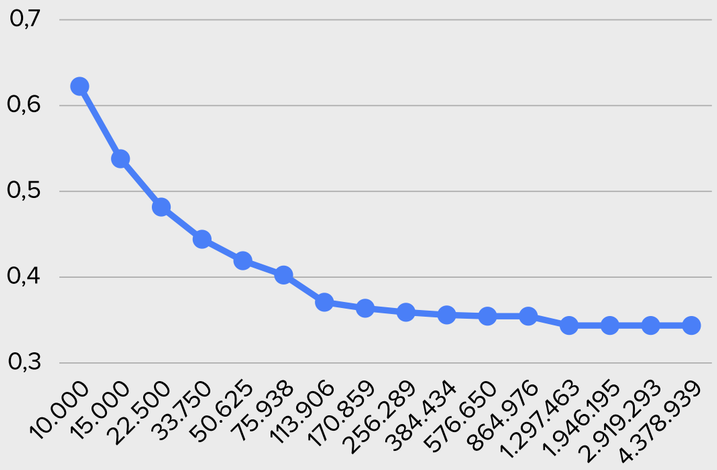


Final mold design: a two-plate mold with the injection point located along the upper area to promote uniform cavity filling

Process cost

Mold Cost	Ctotal_mold	$	3133
Mold Cost per Part	Cmold/part	$	0,006
Processing Cost per Part	Cprocess/part	$	0,284
Material Cost per Part	Cmaterial/part	$	0,048
Yield	0,95		
Total part cost	Cpart	$	**0,345**

Production Volume VS Cost

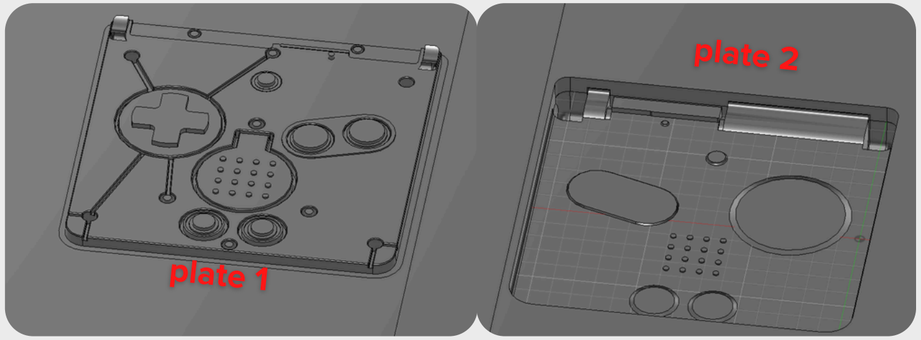


Production Volume	Part Cost
10000	0,622451
50625	0,419172
75938	0,402493
113906	0,370775
576650	0,354506
864976	0,354506
1297463	**0,343654**
1946195	0,343654
2919293	0,343654

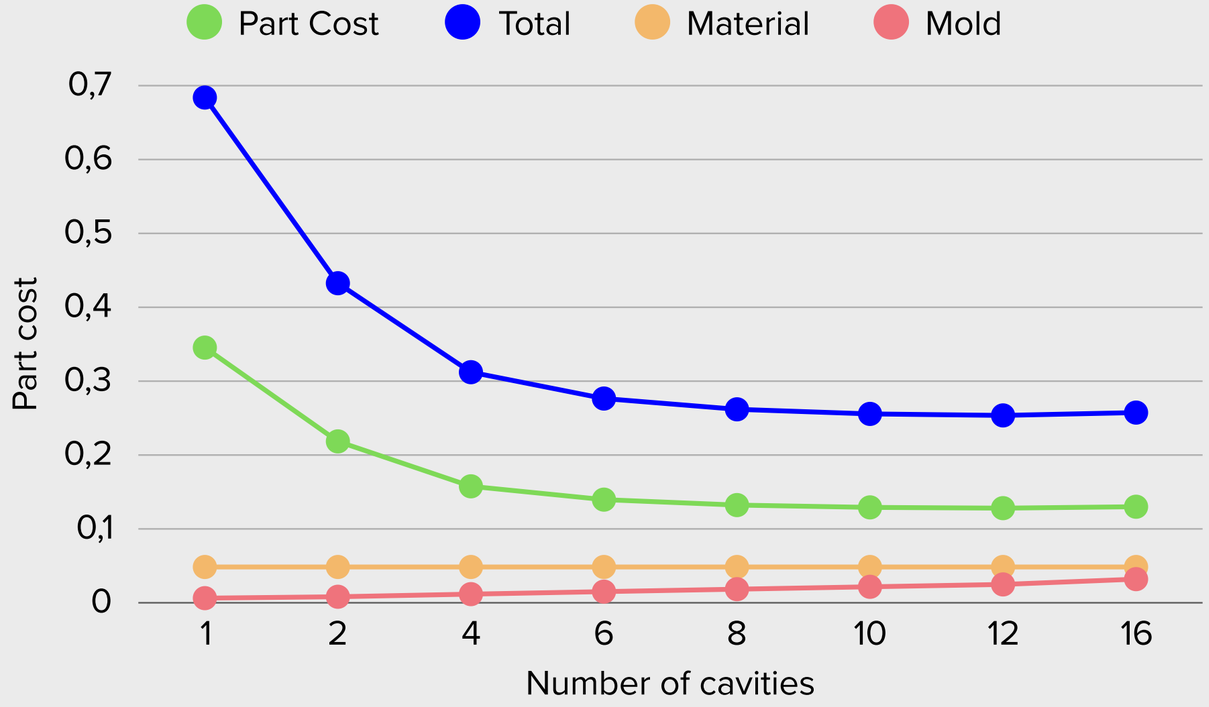
Optimum cost at 1.297.463 capacity and above

Injection molding simulation

2 Plate Mold Design



Production cost vs Number of cavities



Cavities	Total
1	0,683882
2	0,432548
4	0,312133
6	0,276458
8	0,261866
10	0,255657
12	**0,253608**
16	0,257441

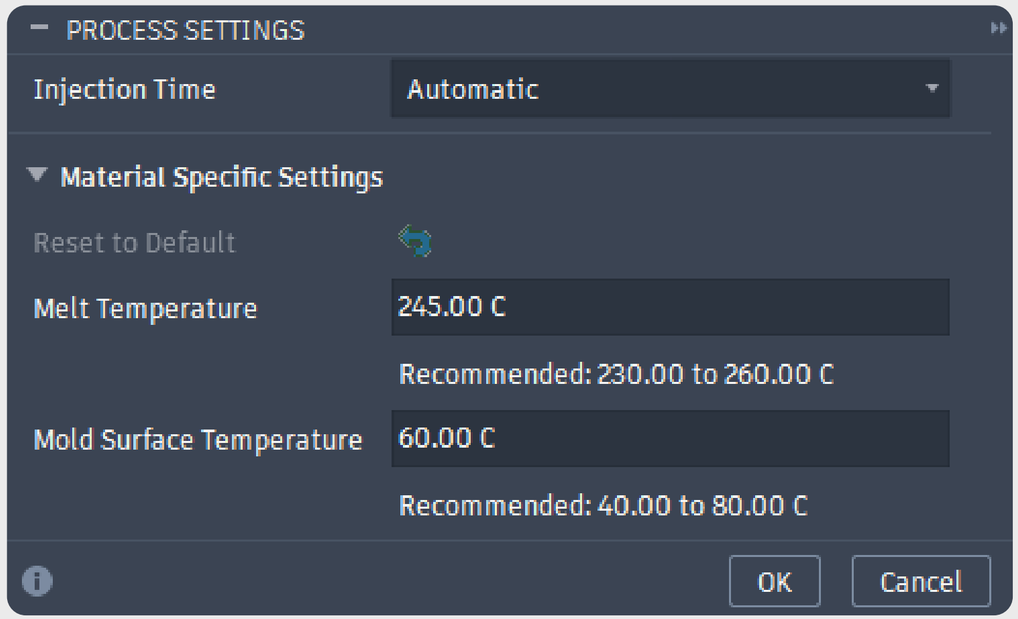
Minimum cost at 12 cavities

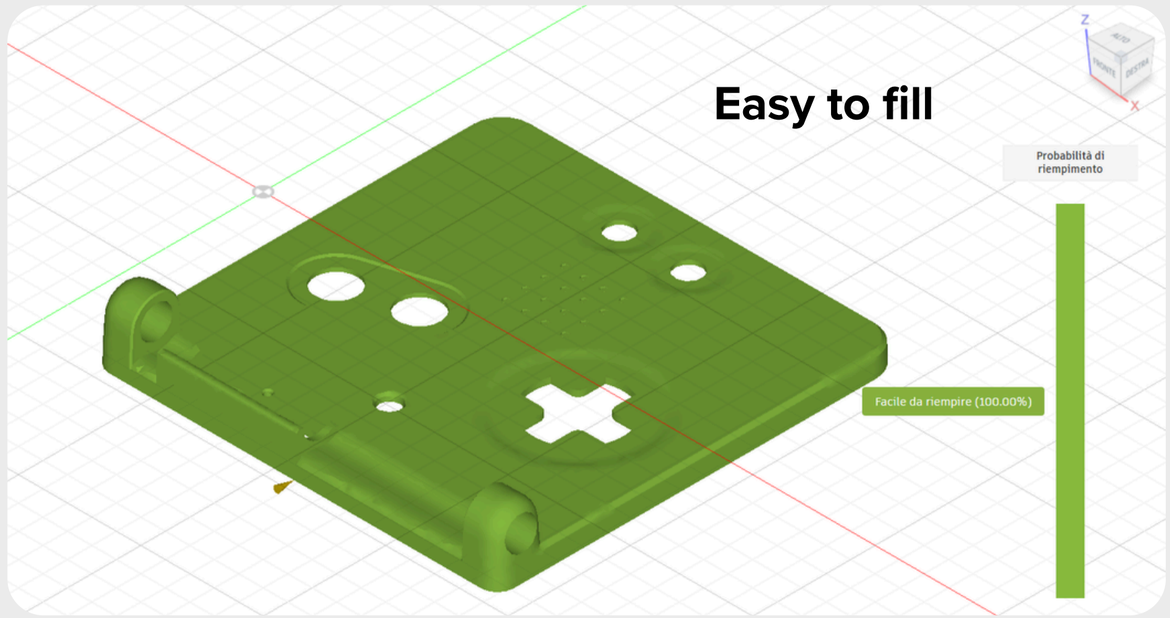
Final mold cost: 12.418$
Final part cost: 0,128$/part
Final material cost: 0,048$/part

Industrialization lab

Injection molding simulation

01 **First simulation**: automatic settings





Injection time t=0.74 s

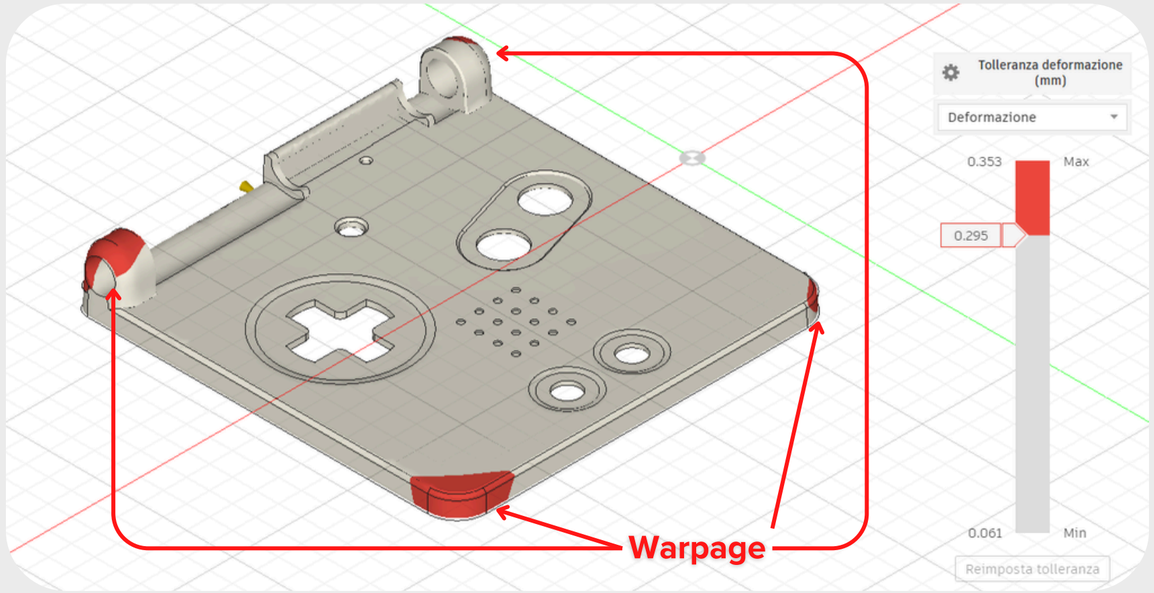
We tried the automatic setting for the first attempt at the Injection molding simulation with the parameters above

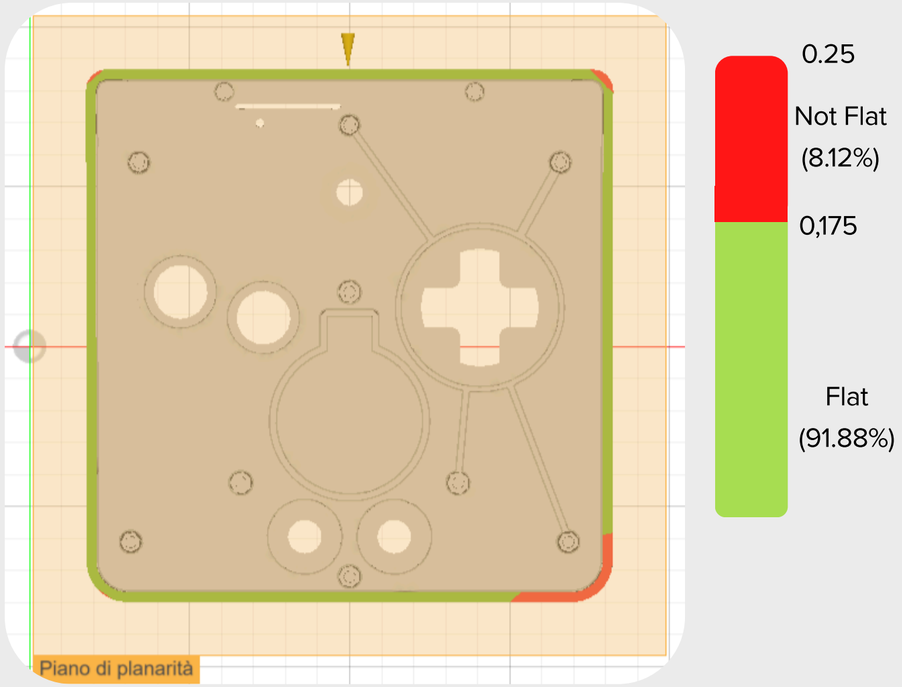
Industrialization lab

Injection molding simulation

01 **First simulation**: automatic settings

Too high planar displacement/warpage





Main issue: a not flat surface prevents proper assembly,

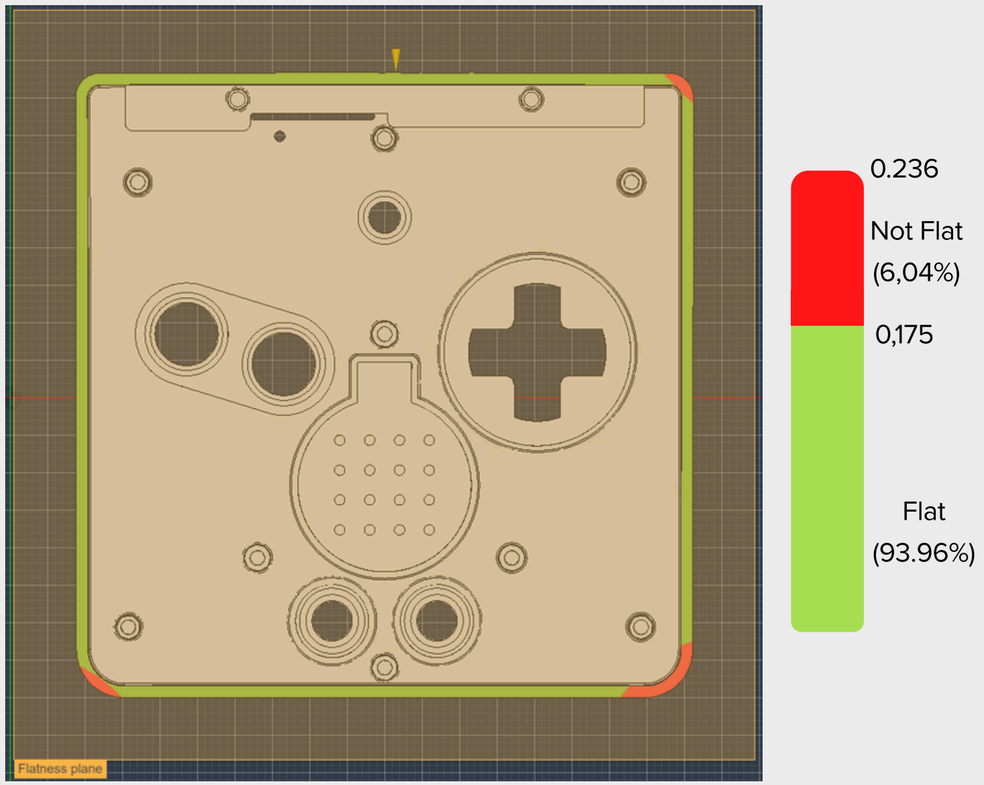
Minor issue: warpage, up to 0.3mm is not optimal but could be accepted

Industrialization lab

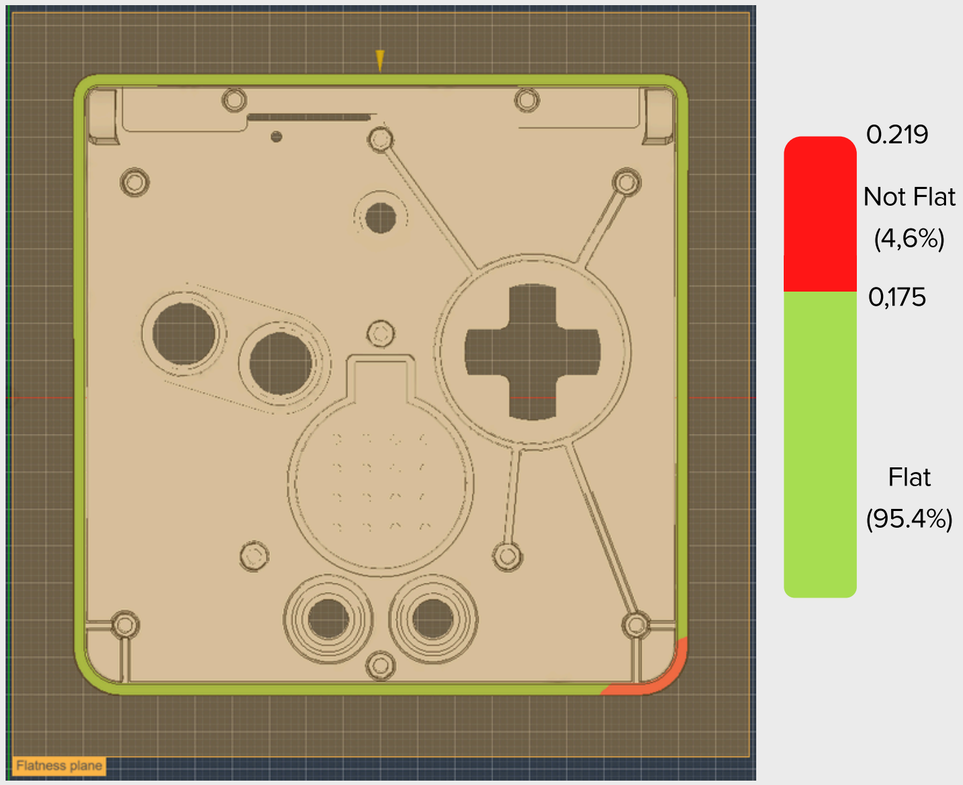
Additional side ribs

Mold temperature: T=60°C (avg)
Melting temperature: T=245°C (avg)
Injection time: t=0.63 s (automatic)

No ribs



With ribs



Industrialization lab

Changing temperatures

Mold temperature: T=40°C
Melting temperature: T=230°C
Injection time: automatic

Mold temperature: T=40°C
Melting temperature: T=245°C (avg)
Injection time: automatic

Mold temperature: T=40°C
Melting temperature: T=260°C
Injection time: automatic

Mold temperature: T=60°C (avg)
Melting temperature: T=230°C
Injection time: automatic

Mold temperature: T=60°C (avg)
Melting temperature: T=245°C (avg)
Injection time: automatic

Mold temperature: T=60°C (avg)
Melting temperature: T=260°C
Injection time: automatic

Mold temperature: T=80°C
Melting temperature: T=230°C
Injection time: automatic

Mold temperature: T=80°C
Melting temperature: T=245°C (avg)
Injection time: automatic

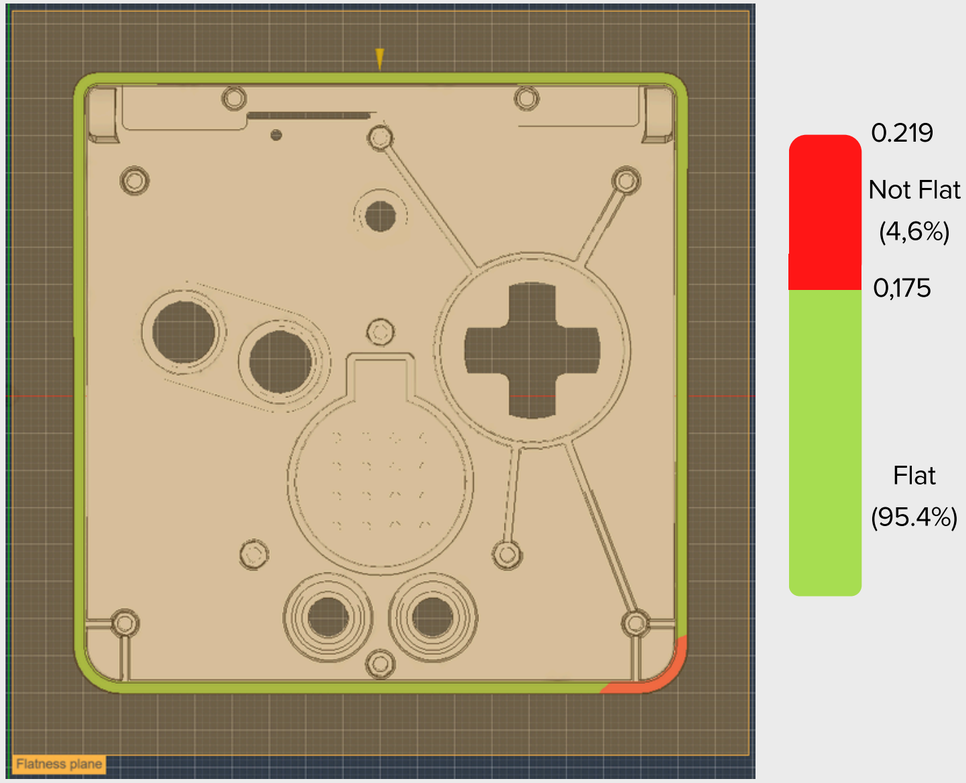
Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: automatic

Industrialization lab

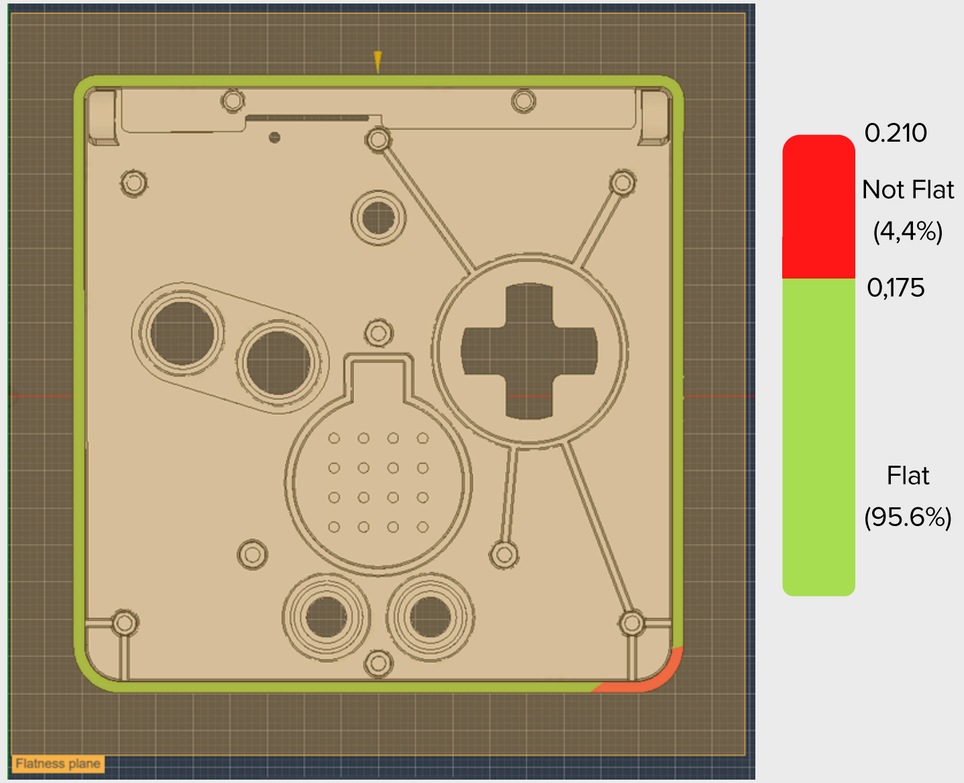
Changing temperatures

Mold temperature: T=60°C
Melting temperature: T=245°C
Injection time: t=0.63 s (automatic)

Best previous results

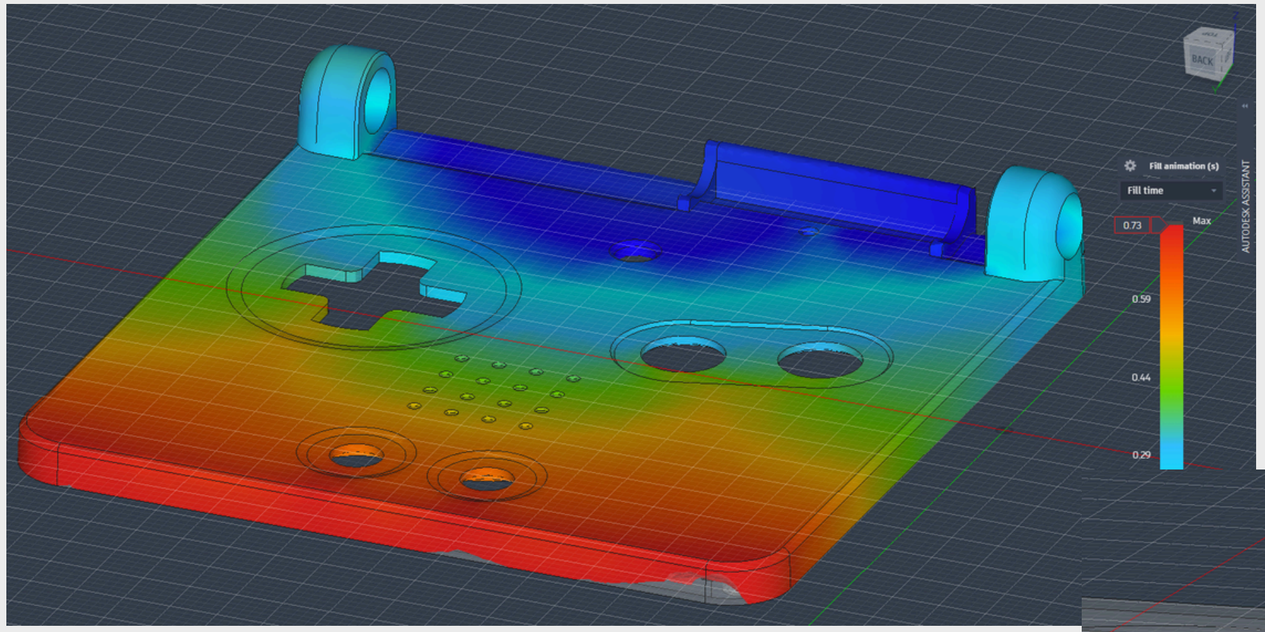


Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: t=0.73 s (automatic)

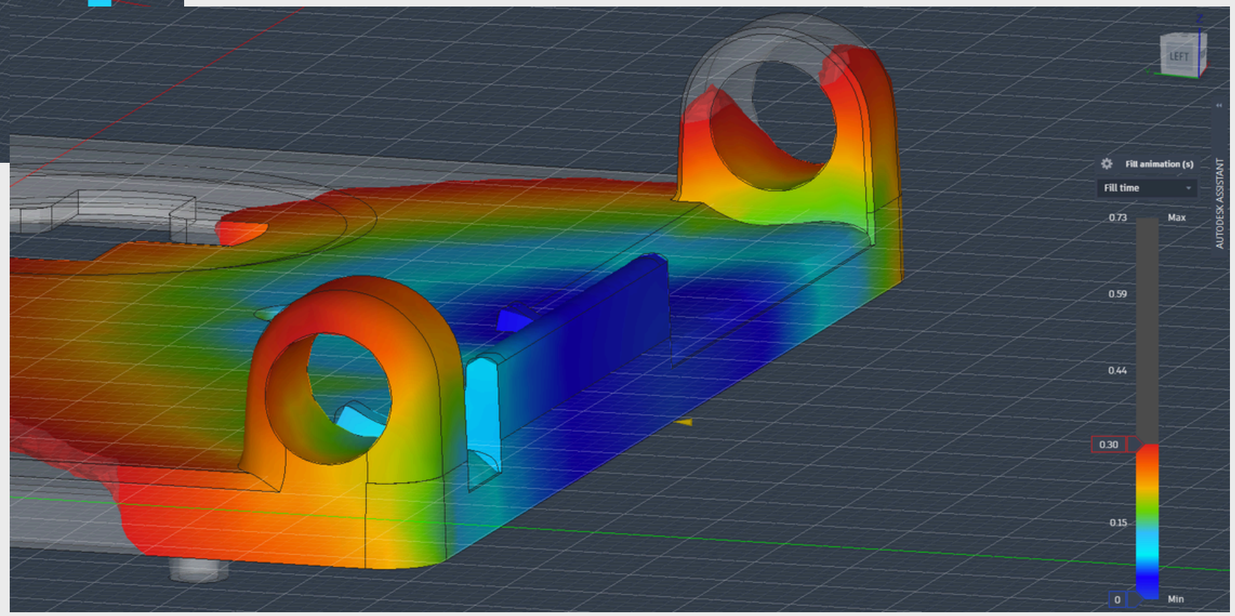


Industrialization lab

Changing injection point



Material flow simulation with injection point in the middle of the component

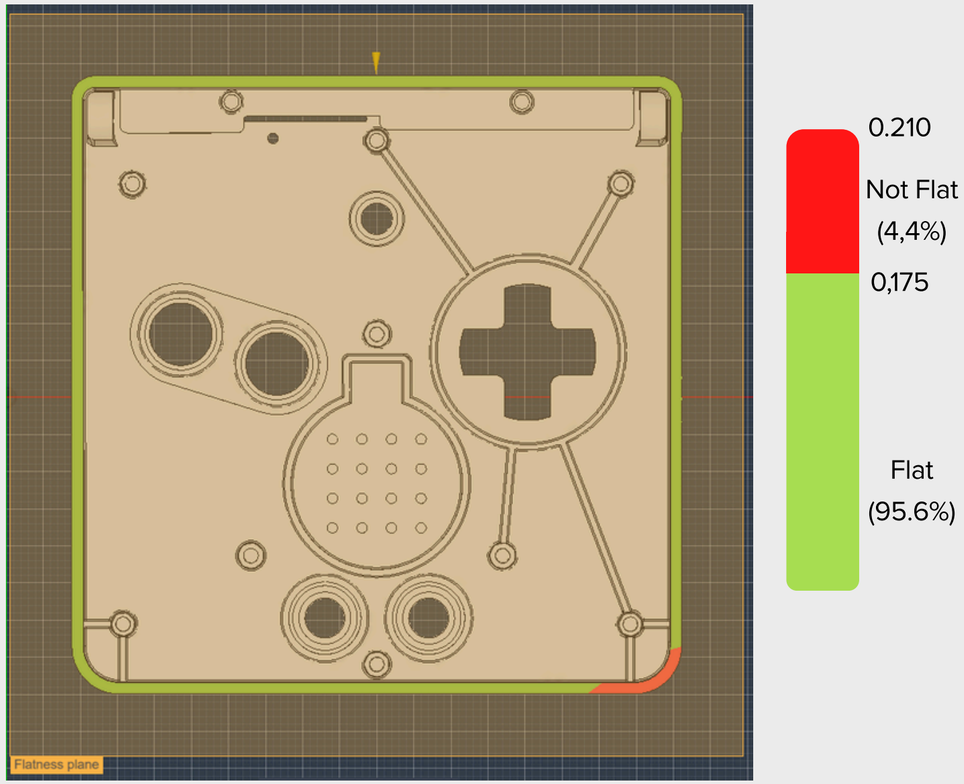


In order to reduce the warpage we will try to slightly move the injection point in order to promote symmetry

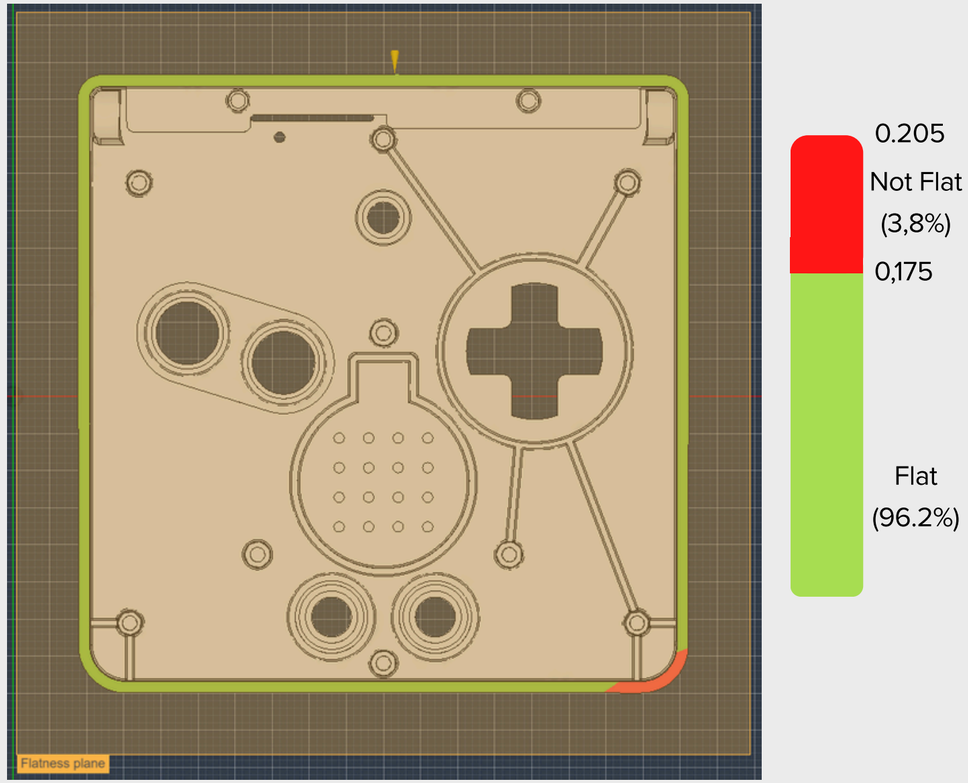
Industrialization lab

Changing injection point

Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: t=0.73 s (automatic)



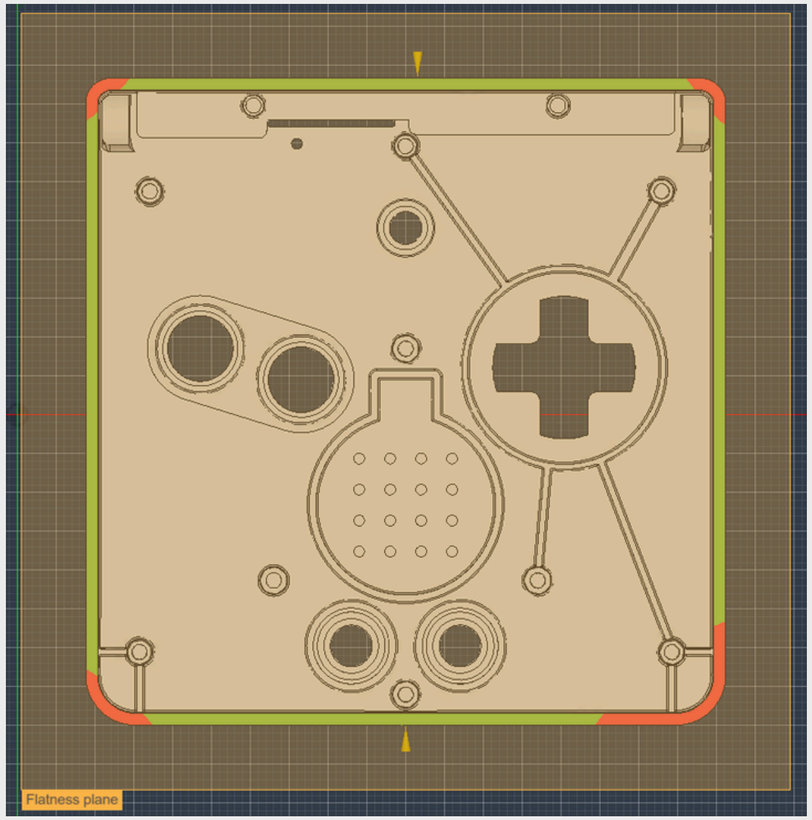
Moving the injection point slightly to the right to get closer to the region that gets filled last

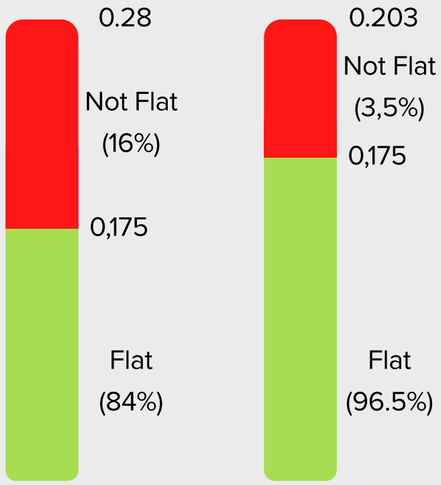


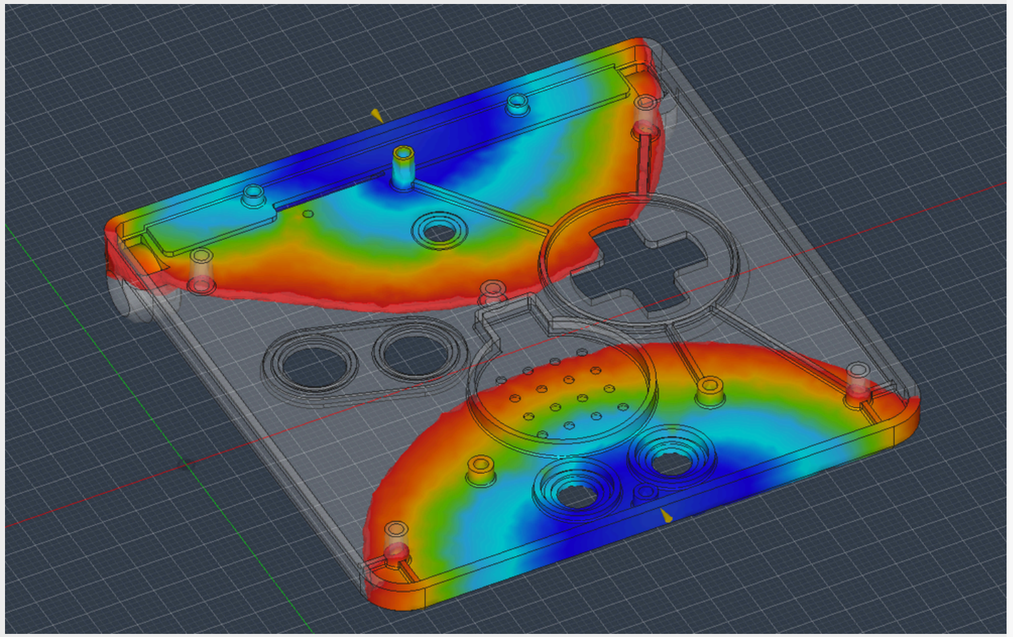
Industrialization lab

Testing with 2 injectors

Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: t=0.41s (automatic)
Two injection points





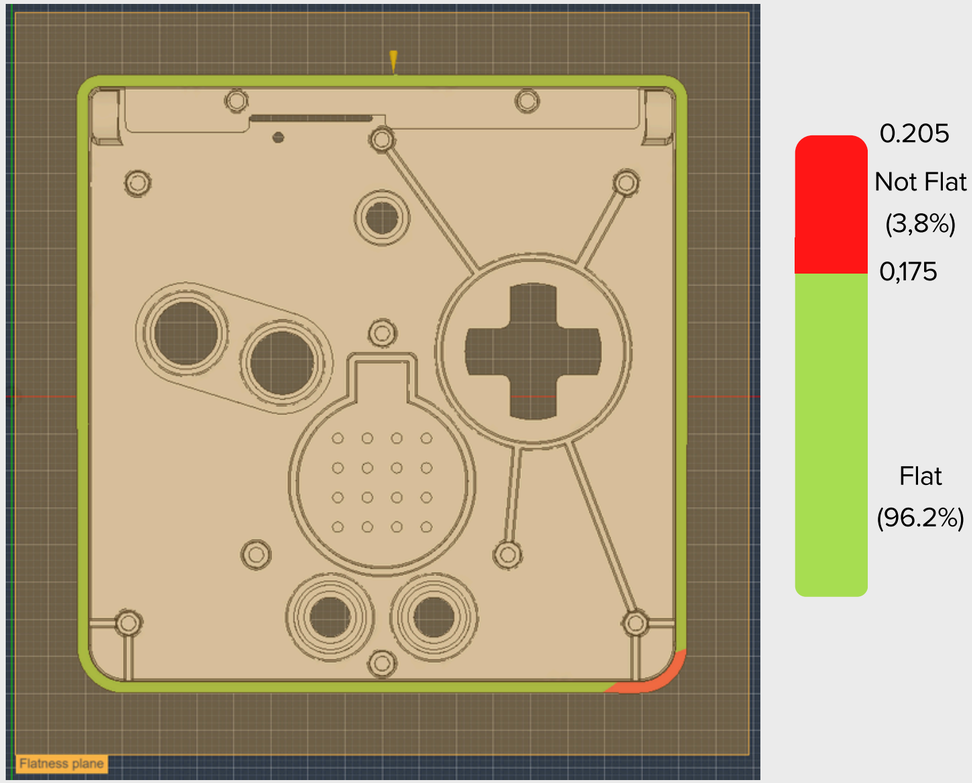


Worse than previous results

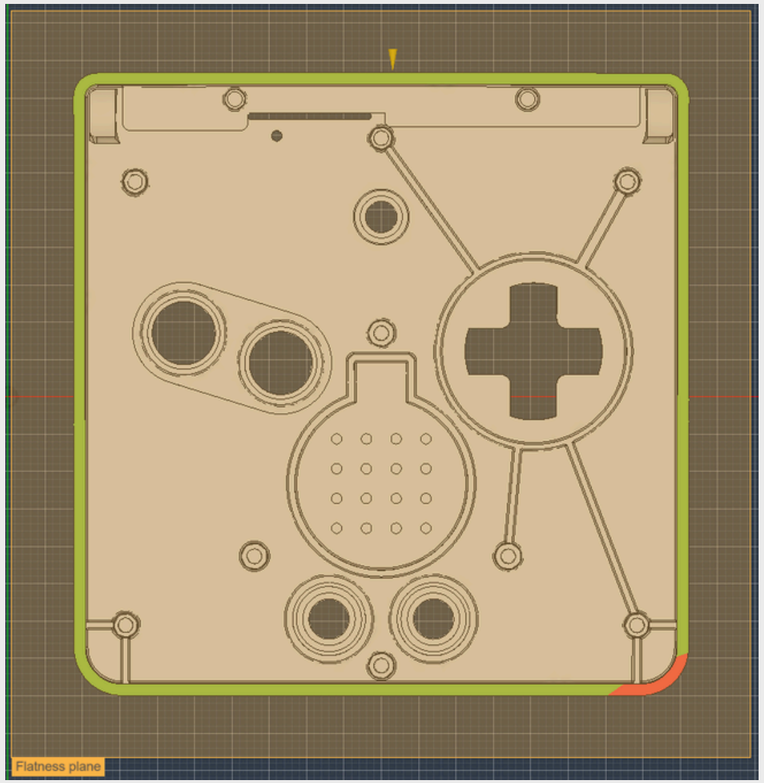
Industrialization lab

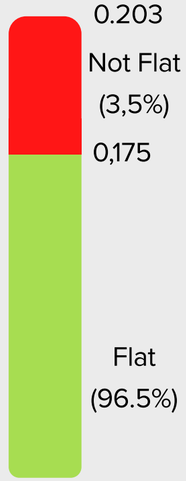
Changing injection time

Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: t=0.73s (automatic)
New injection point



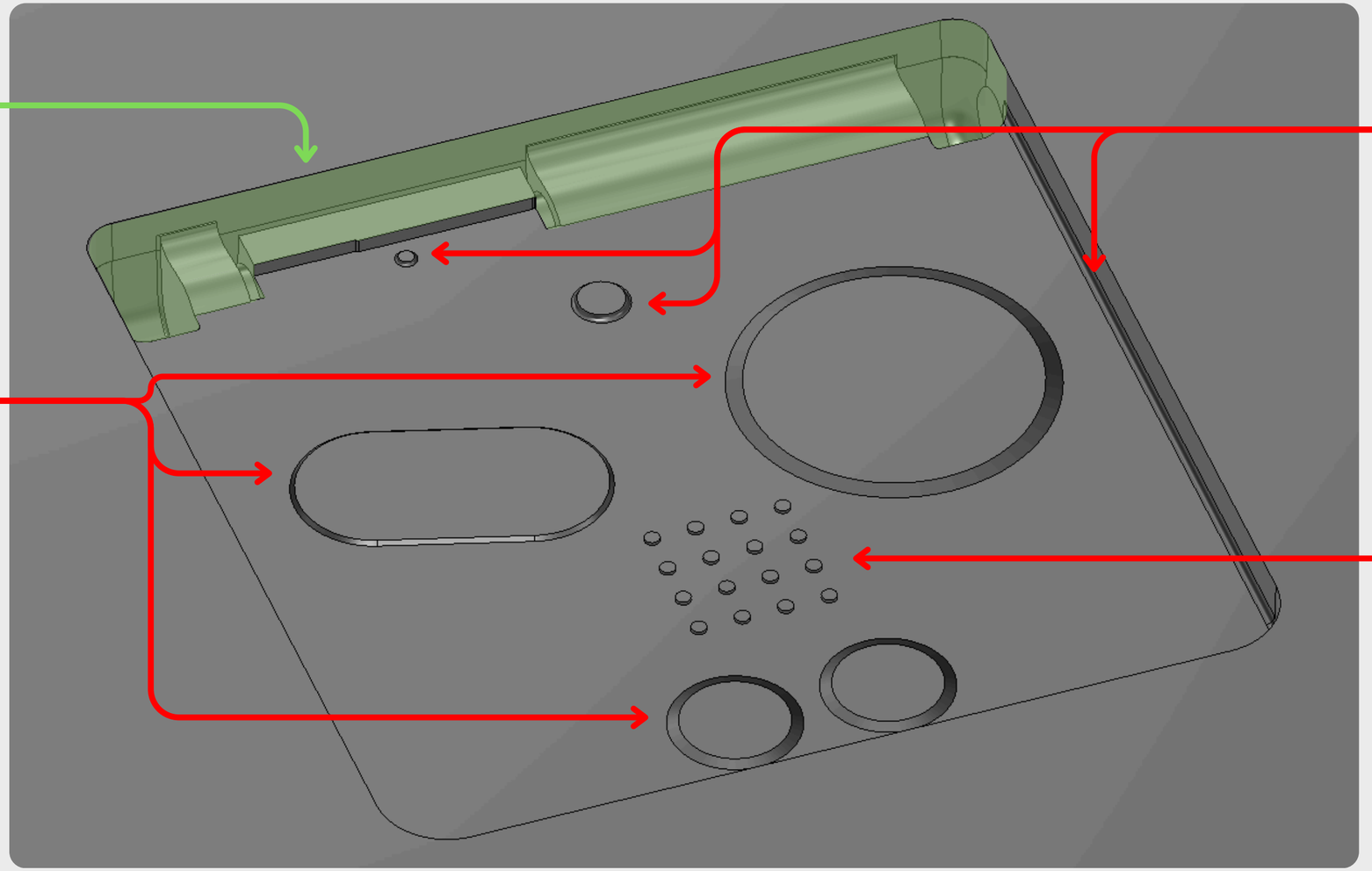
Mold temperature: T=80°C
Melting temperature: T=260°C
Injection time: t=0.63s (specified)
New injection point





Manufacturing lab

Mold manufacturing process

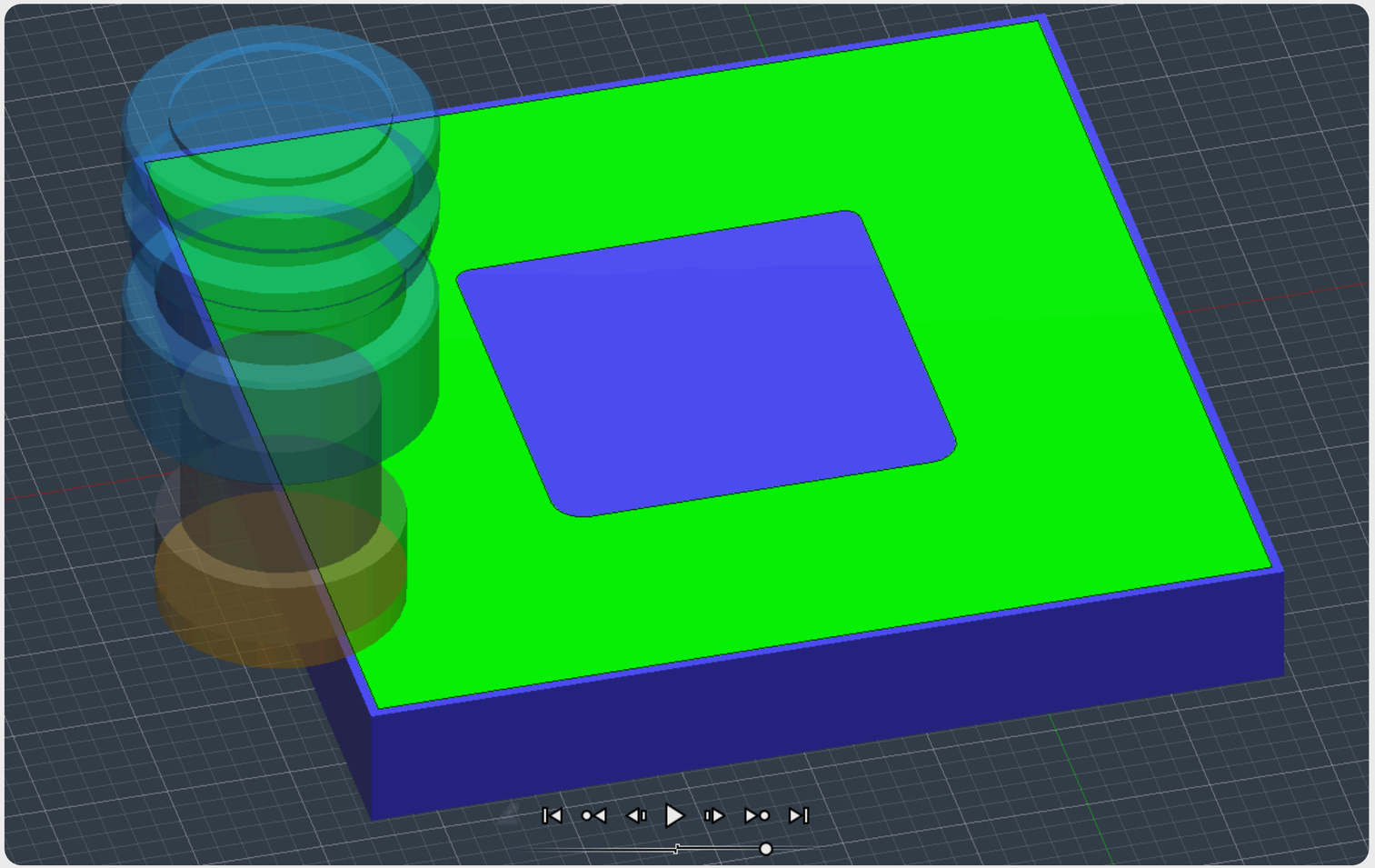


Manufacturing lab

Manufacturing process

Step 1

Face milling



Stock simulation

Positive Material

Negative Material

∅50mm Face Mill

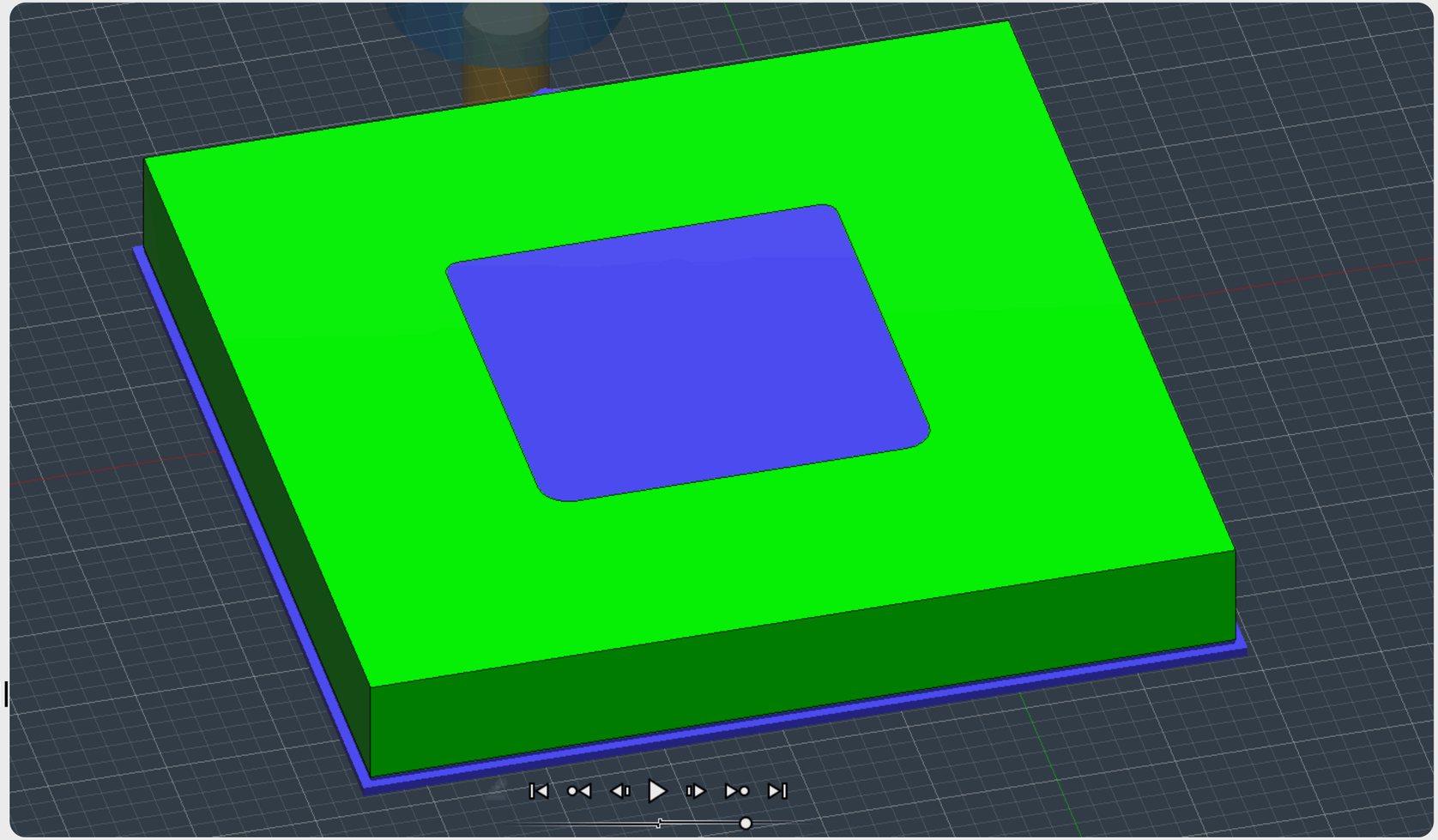
t = 1:31min

Manufacturing lab

Manufacturing process

Step 2

2D Contour



Stock simulation

Positive Material

Negative Material

∅18mm Flat Endmill

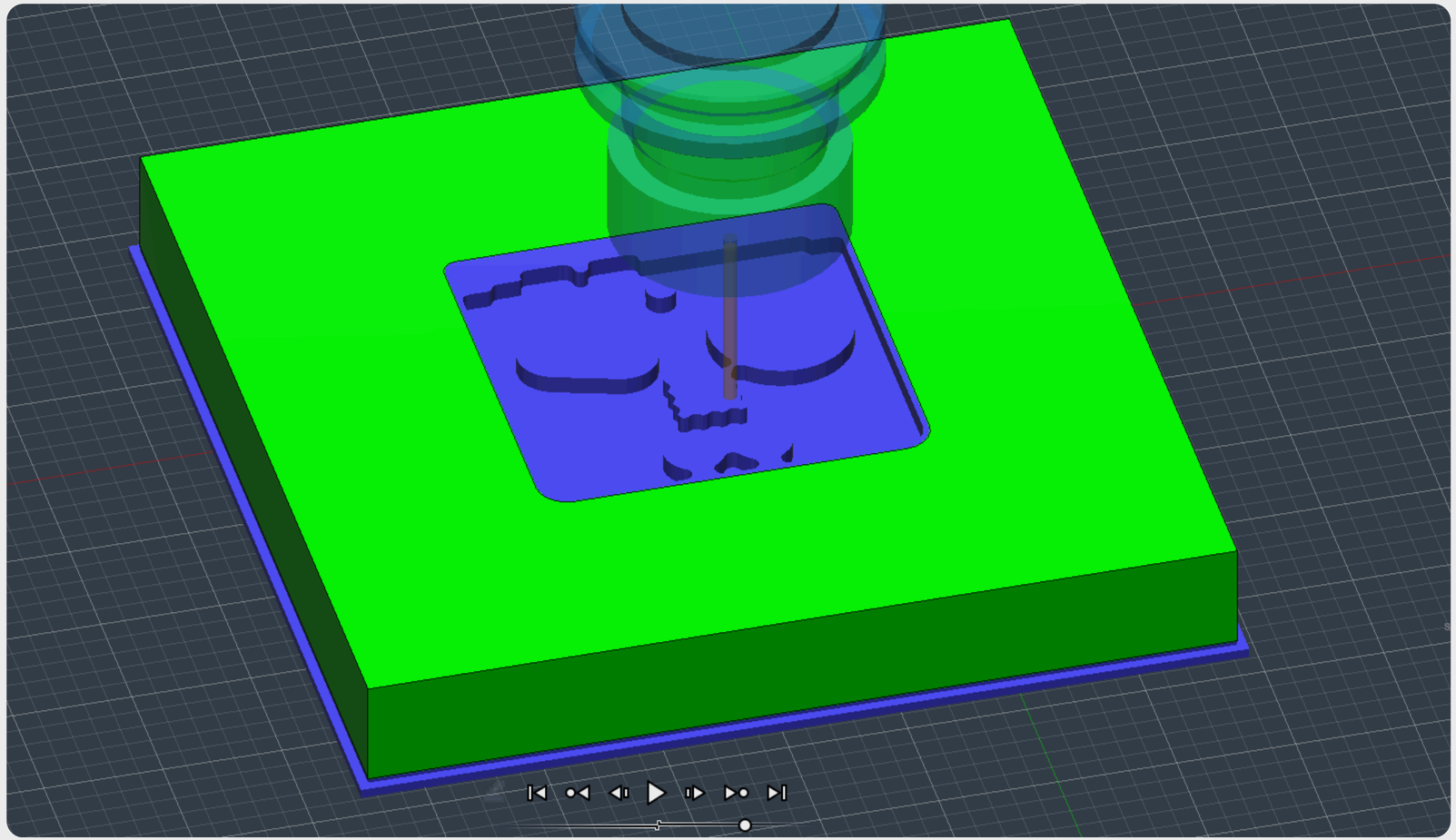
t = 53s

Step 3

2D Adaptive

∅2,5mm Flat Endmill

t = 4:29min



Stock simulation

Positive Material

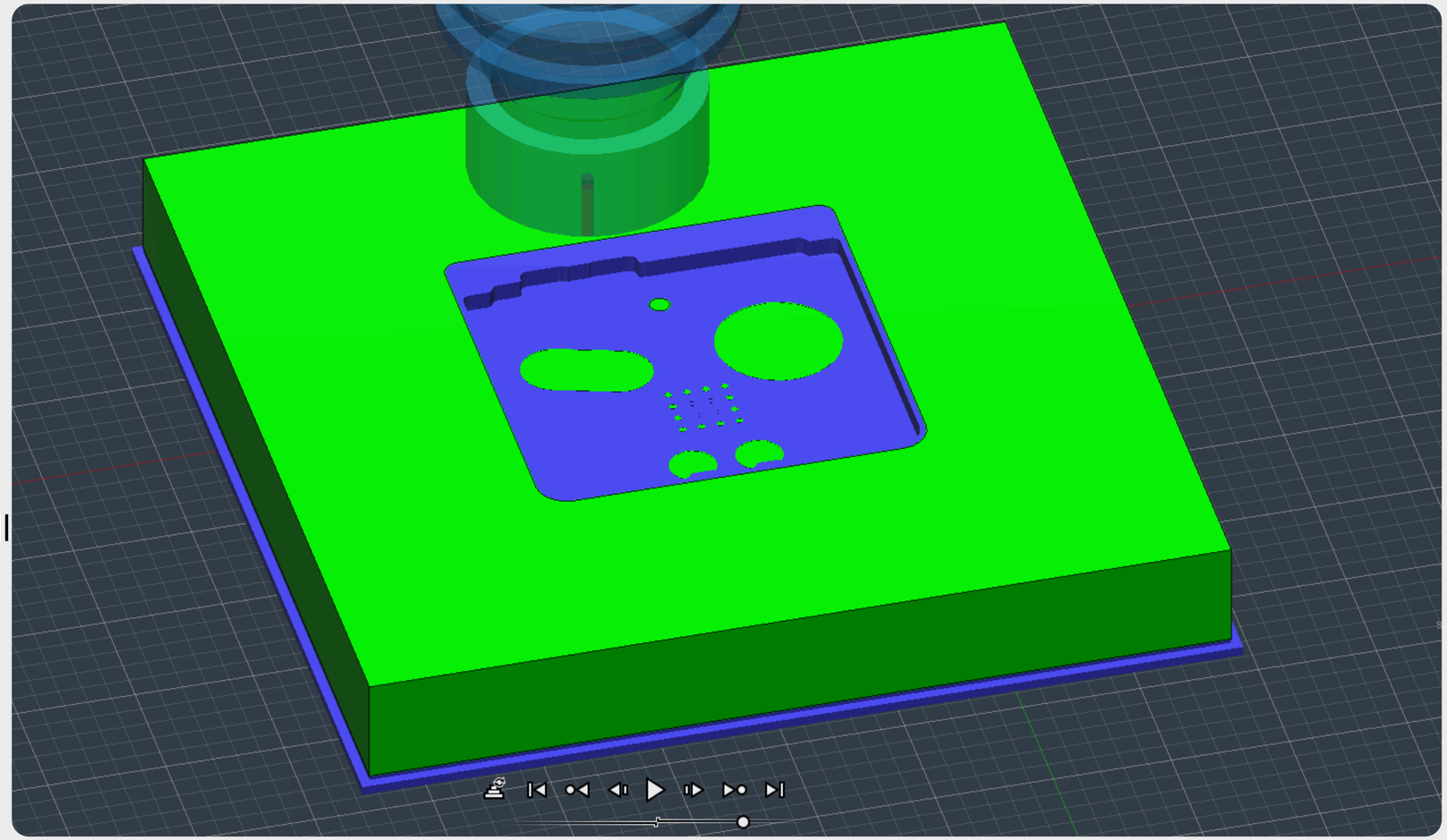
Negative Material

Step 4

Face milling

∅2,5mm Flat Endmill

t = 6:07min

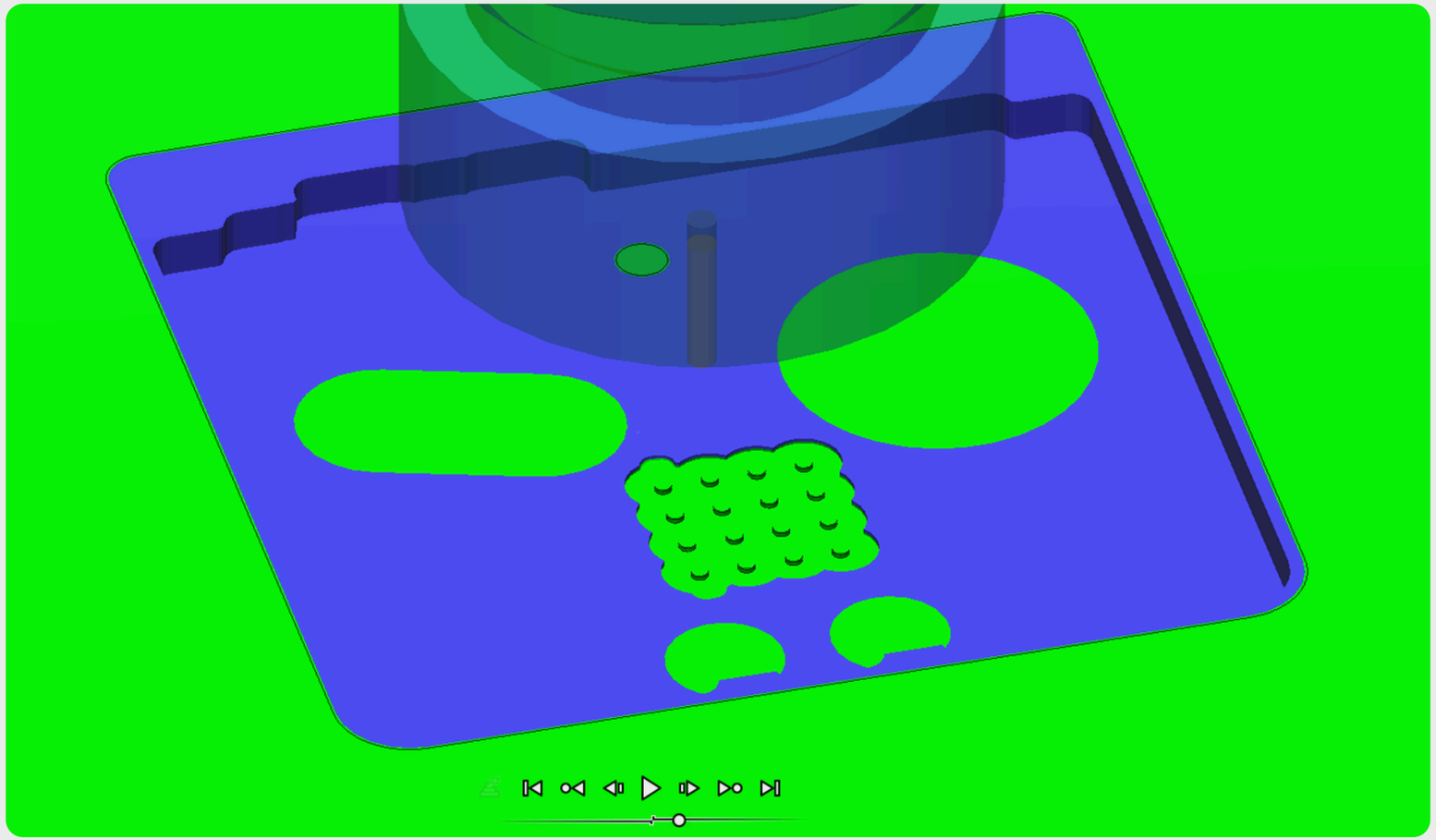


Step 5

2D Contour

∅2,5mm Flat Endmill

t = 29s



Stock simulation

Positive Material

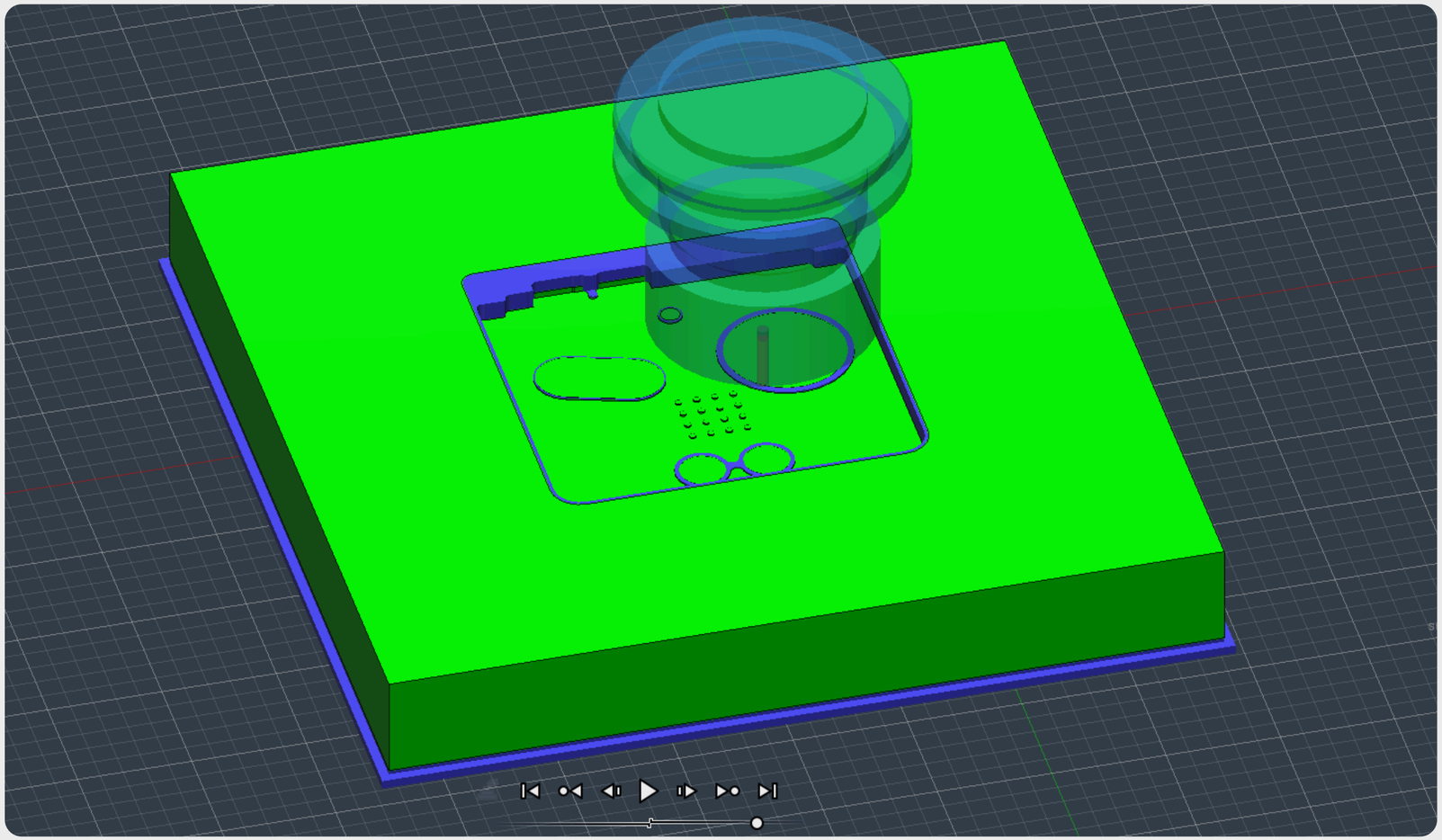
Negative Material

Step 6

2D Adaptive

∅2,5mm Flat Endmill

t = 7:15min



Stock simulation

Positive Material

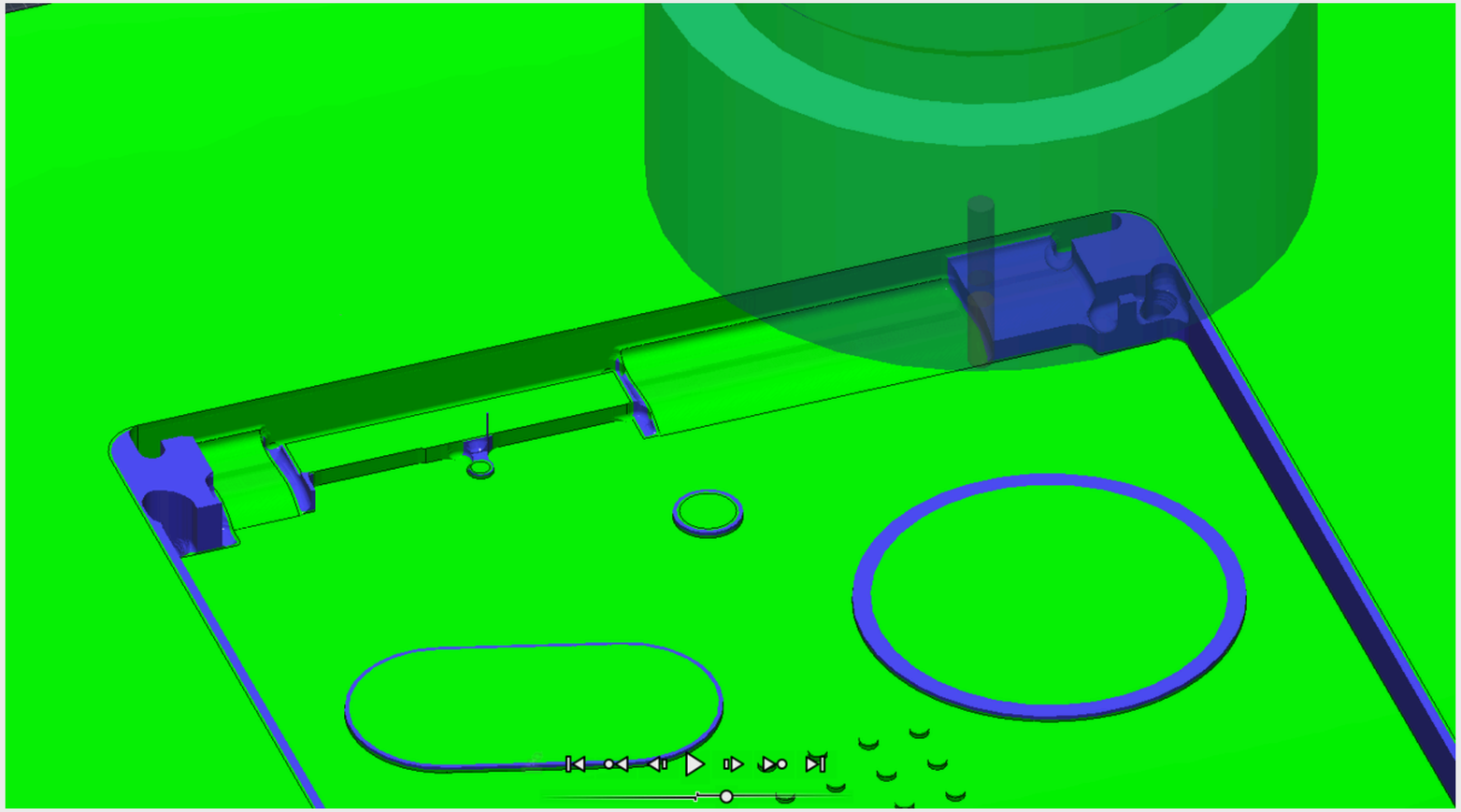
Negative Material

Step 7

3D Parallel

∅2mm Bullnose Endmill

t = 33:31min

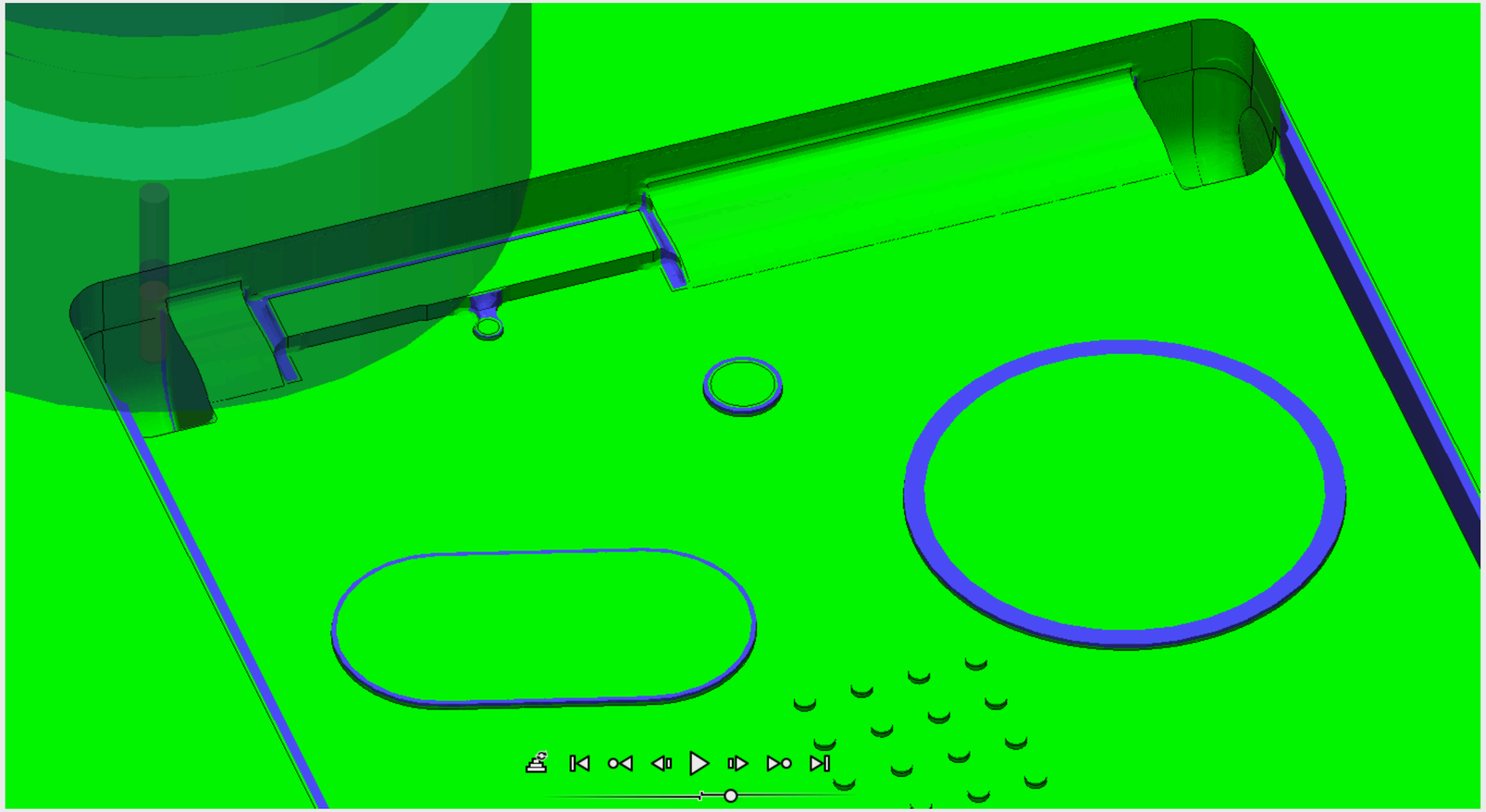


Step 8

3D Parallel

∅2mm Bullnose Endmill

t = 25:04min



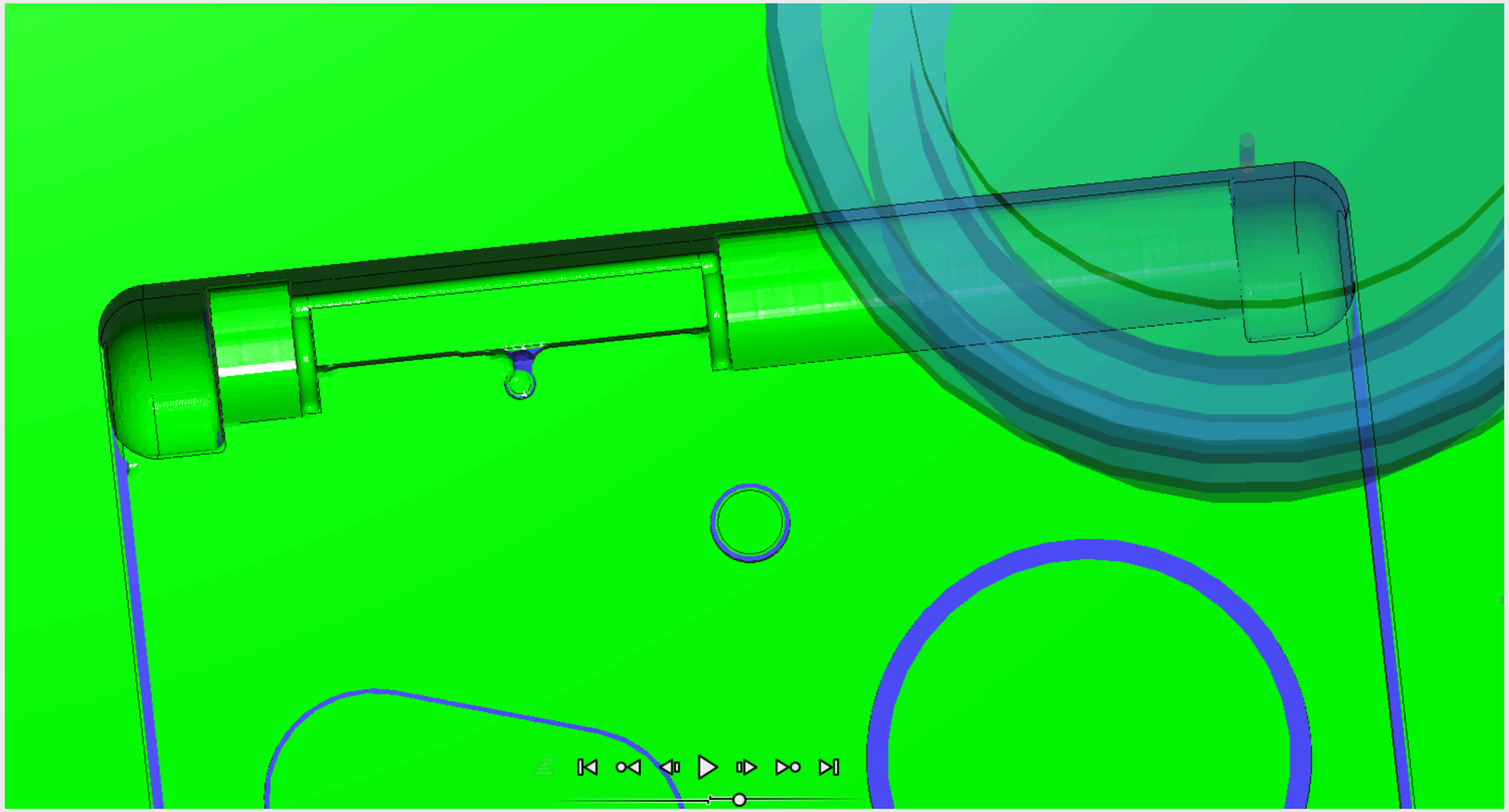
Manufacturing lab

Step 9

3D Parallel

∅2mm Bullnose Endmill

t = 9:42min



Stock simulation

Positive Material

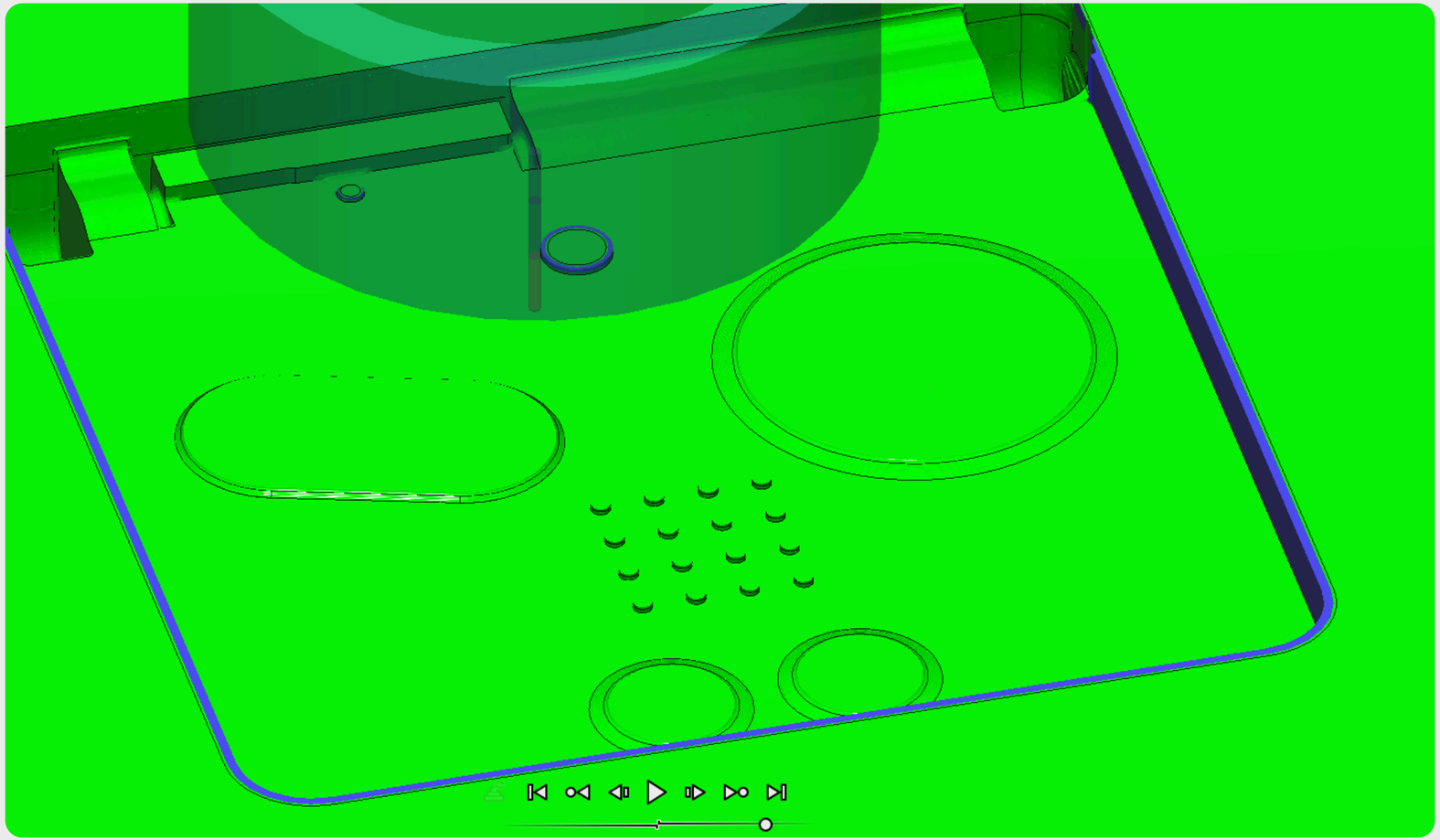
Negative Material

Step 10

3D Scallop

∅1mm Bullnose Endmill

t = 3:02min

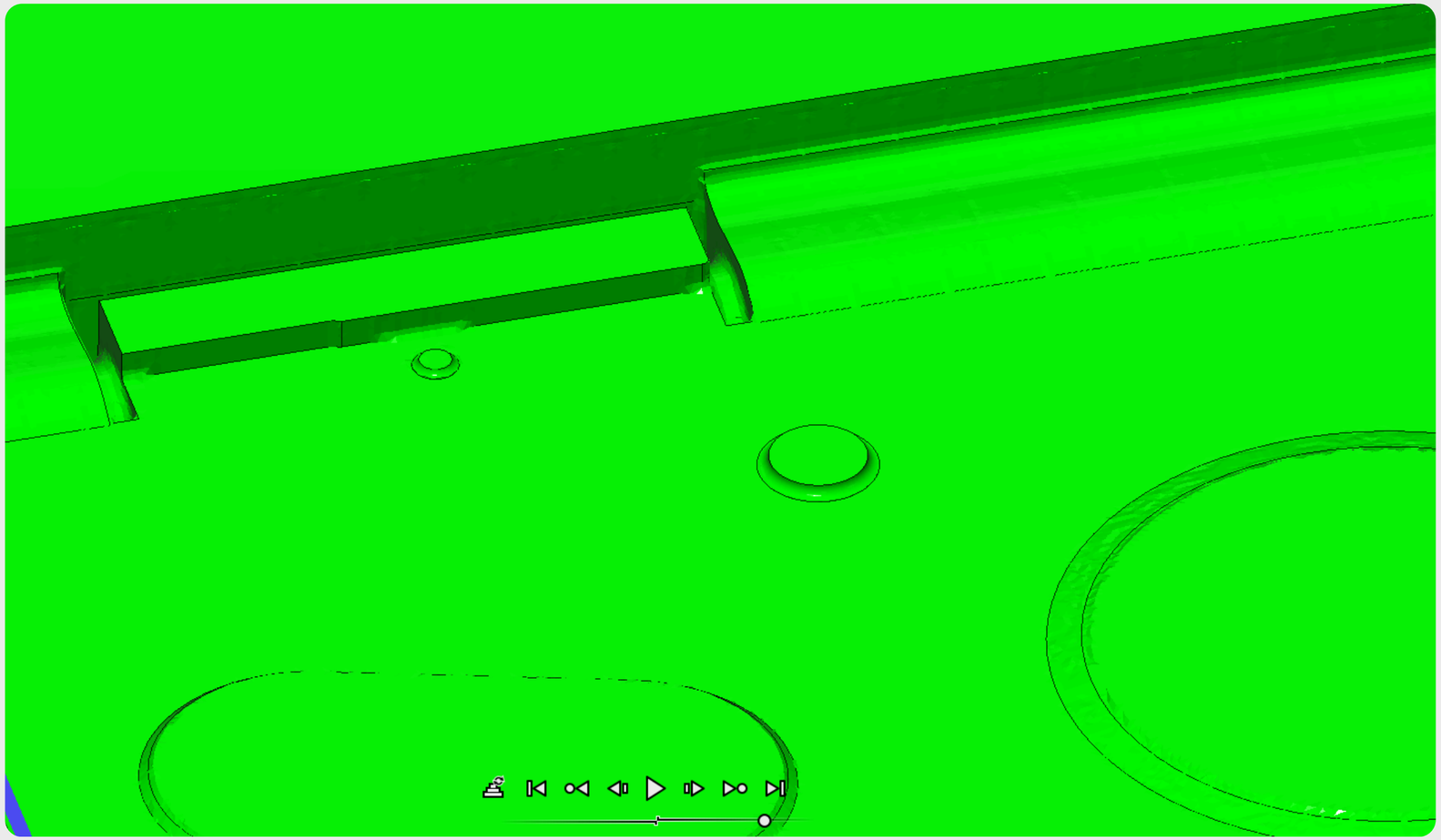


Step 11

3D Ramp

∅1mm Bullnose Endmill

t = 8s

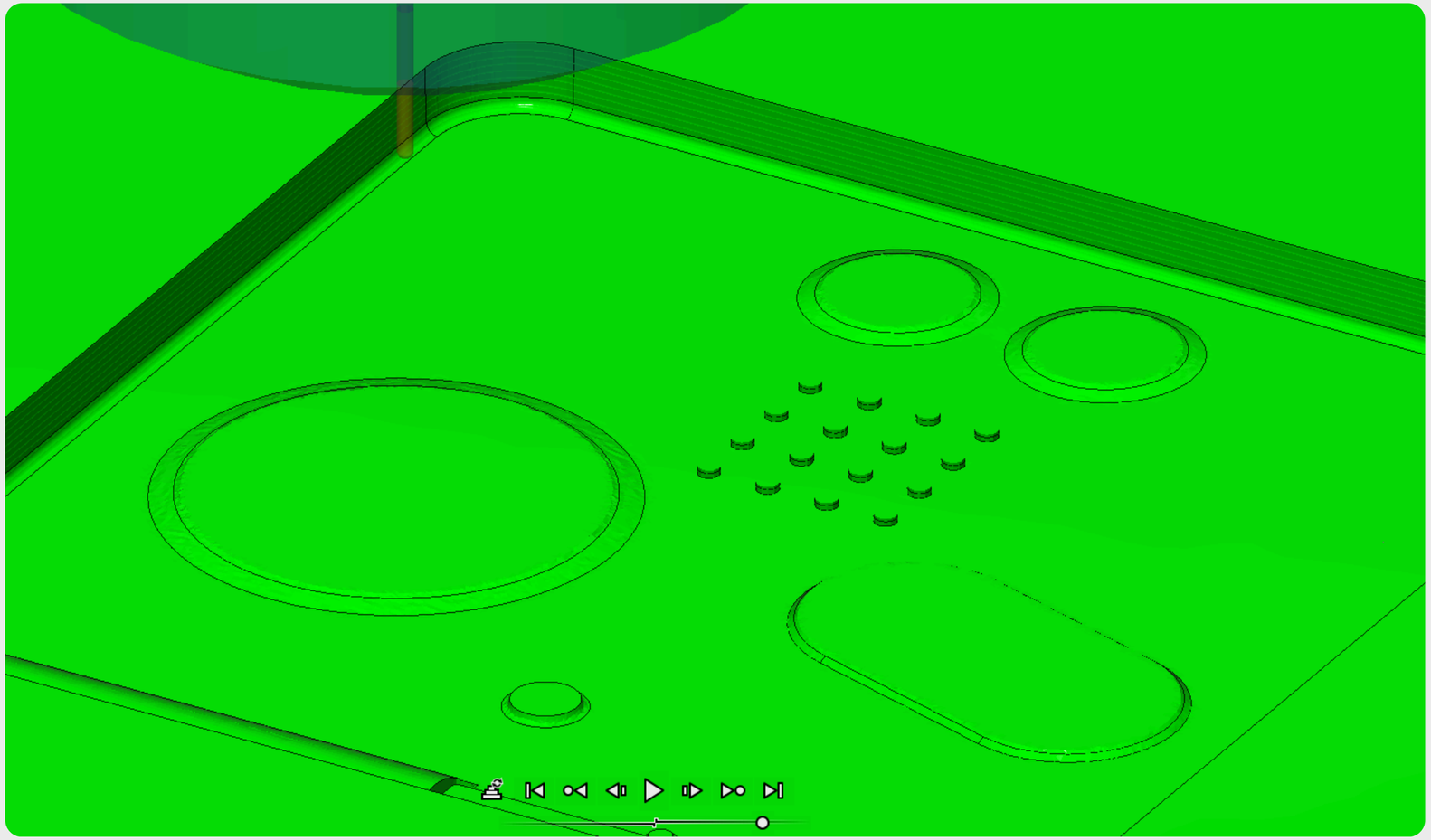


Step 12

3D Flow

∅1mm Bullnose Endmill

t = 2:53min



Step 13

2D Contour

∅1mm Flat Endmill

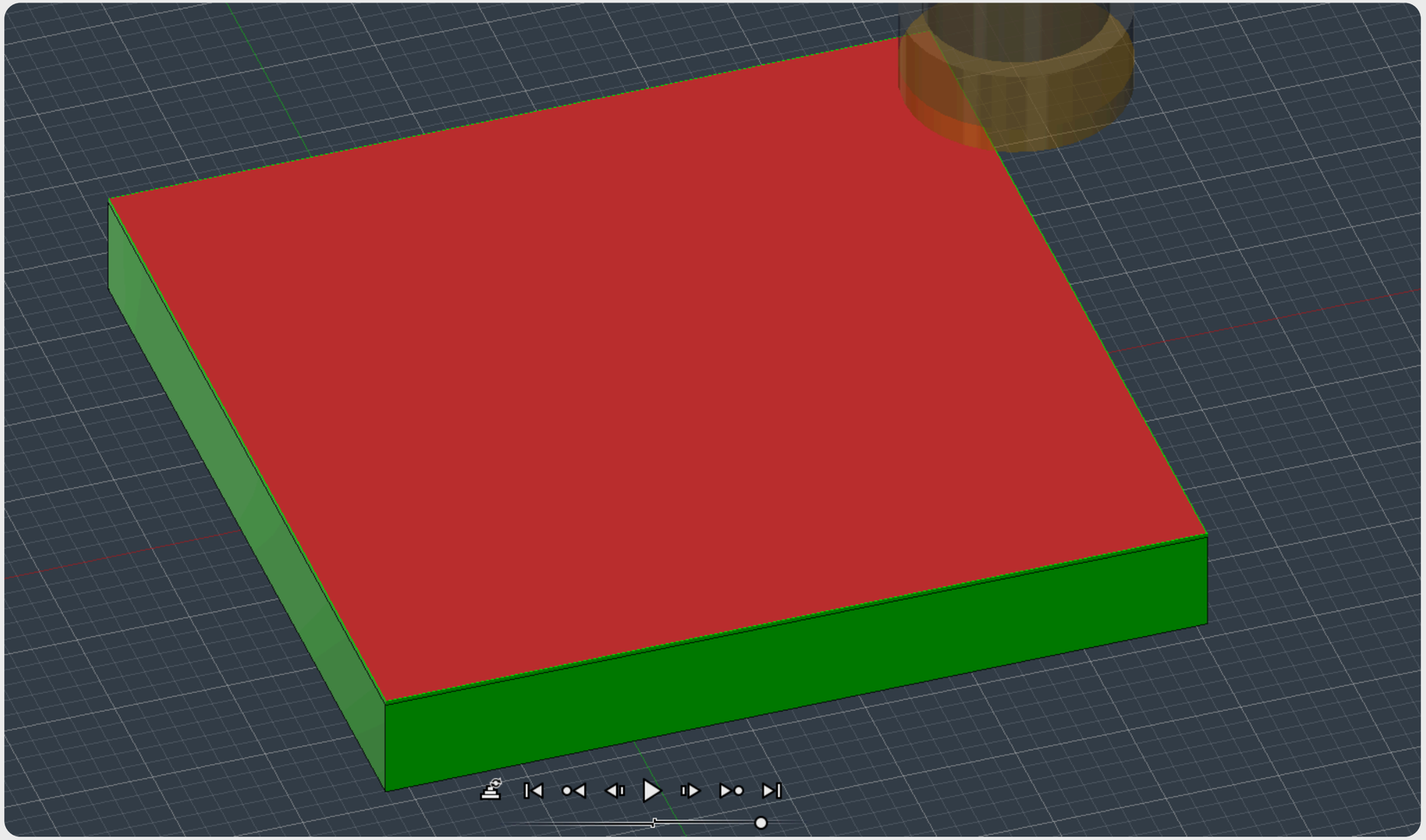
t = 1s

Step 14

Face milling

∅50mm Face Mill

t = 3:23min



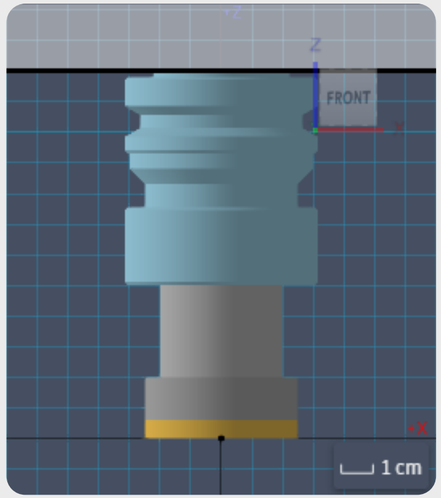
Stock simulation

Positive Material

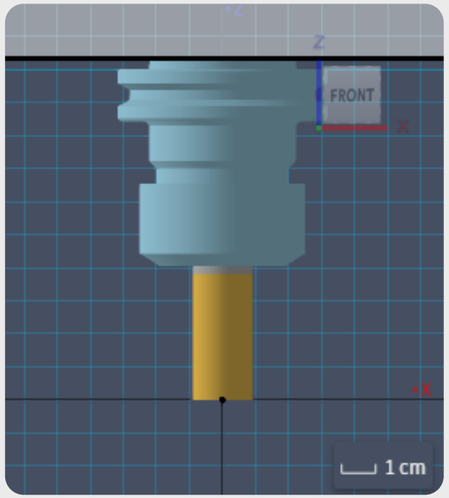
Negative Material

Recap manufacturing process

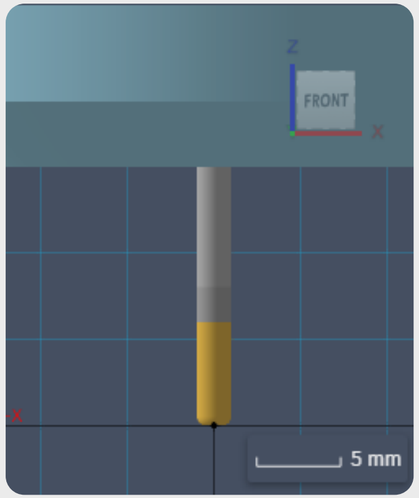
∅50mm Face Mill



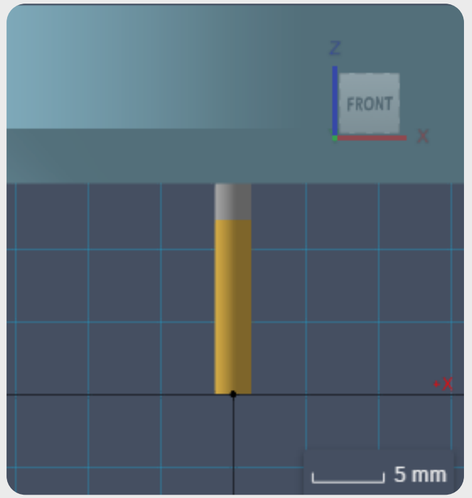
∅18mm Flat Endmill



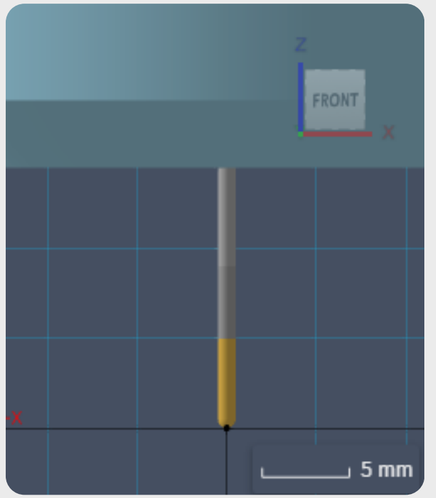
∅2mm Bullnose Endmill



∅2,5mm Flat Endmill



∅1mm Bullnose Endmill

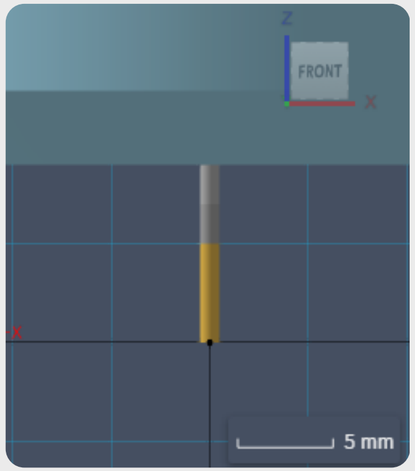


Total time

t = 1h 38min 28s

Optional:

∅1mm Flat Endmill



Thank You

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